

Tabla de Integrales indefinidas

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REGLAS PARA UNA INTEGRACION

$$1 \quad \int a \, dx = ax$$

$$2 \quad \int af(x) \, dx = a \int f(x) \, dx$$

$$3 \quad \int (u \pm v \pm w \pm \dots) \, dx = \int u \, dx \pm \int v \, dx \pm \int w \, dx \pm \dots$$

$$4 \quad \int u \, dv = uv - \int v \, du$$

$$5 \quad \int f(ax) \, dx = \frac{1}{a} \int f(u) \, du$$

$$6 \quad \int F\{f(x)\} \, dx = \int F(u) \frac{dx}{du} \, du = \int \frac{F(u)}{f'(x)} \, du$$

$$7 \quad \int u^n \, du = \frac{u^{n+1}}{n+1}, \quad n \neq -1 \quad \text{Para } n = -1, \text{ véase } 8$$

$$8 \quad \int \frac{du}{u} = \ln u \quad \text{si } u > 0 \quad \text{o} \quad \ln(-u) \quad \text{si } u < 0 \\ = \ln |u|$$

$$9 \quad \int e^u \, du = e^u$$

$$10 \quad \int a^u \, du = \int e^{u \ln a} \, du = \frac{e^{u \ln a}}{\ln a} = \frac{a^u}{\ln a}, \quad a > 0, \quad a \neq 1$$

NOTA: a, b, p, q, n
son constantes

u, v, w son funciones
de x

$$11 \quad \int \operatorname{sen} u \, du = -\cos u$$

$$12 \quad \int \cos u \, du = \operatorname{sen} u$$

$$13 \quad \int \tan u \, du = \ln \sec u = -\ln \cos u$$

$$14 \quad \int \cot u \, du = \ln \operatorname{sen} u$$

$$15 \quad \int \sec u \, du = \ln (\sec u + \tan u) = \ln \tan \left(\frac{u}{2} + \frac{\pi}{4} \right)$$

$$16 \quad \int \csc u \, du = \ln (\csc u - \cot u) = \ln \tan \frac{u}{2}$$

$$17 \quad \int \sec^2 u \, du = \tan u$$

$$18 \quad \int \csc^2 u \, du = -\cot u$$

$$19 \quad \int \tan^2 u \, du = \tan u - u$$

$$20 \quad \int \cot^2 u \, du = -\cot u - u$$

$$21 \quad \int \operatorname{sen}^2 u \, du = \frac{u}{2} - \frac{\operatorname{sen} 2u}{4} = \frac{1}{2}(u - \operatorname{sen} u \cos u)$$

$$22 \quad \int \cos^2 u \, du = \frac{u}{2} + \frac{\operatorname{sen} 2u}{4} = \frac{1}{2}(u + \operatorname{sen} u \cos u)$$

$$23 \quad \int \sec u \tan u \, du = \sec u$$

$$24 \quad \int \csc u \cot u \, du = -\csc u$$

$$25 \quad \int \sinh u \, du = \cosh u$$

$$26 \quad \int \cosh u \, du = \sinh u$$

$$27 \quad \int \tanh u \, du = \ln \cosh u$$

$$28 \quad \int \coth u \, du = \ln \sinh u$$

$$29 \quad \int \operatorname{sech} u \, du = \operatorname{sen}^{-1}(\tanh u) \quad \text{o} \quad 2 \tan^{-1} e^u$$

$$30 \quad \int \operatorname{csch} u \, du = \ln \tanh \frac{u}{2} \quad \text{o} \quad -\coth^{-1} e^u$$

$$31 \quad \int \operatorname{sech}^2 u \, du = \tanh u$$

$$32 \quad \int \operatorname{csch}^2 u \, du = -\coth u$$

$$33 \quad \int \tanh^2 u \, du = u - \tanh u$$

- 34 $\int \coth^2 u \, du = u - \coth u$
- 35 $\int \sinh^2 u \, du = \frac{\sinh 2u}{4} - \frac{u}{2} = \frac{1}{2}(\sinh u \cosh u - u)$
- 36 $\int \cosh^2 u \, du = \frac{\sinh 2u}{4} + \frac{u}{2} = \frac{1}{2}(\sinh u \cosh u + u)$
- 37 $\int \operatorname{sech} u \tanh u \, du = -\operatorname{sech} u$
- 38 $\int \operatorname{csch} u \coth u \, du = -\operatorname{csch} u$
- 39 $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{u}{a}$
- 40 $\int \frac{du}{u^2 - a^2} = \frac{1}{2a} \ln \left(\frac{u-a}{u+a} \right) = -\frac{1}{a} \coth^{-1} \frac{u}{a} \quad u^2 > a^2$
- 41 $\int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln \left(\frac{a+u}{a-u} \right) = \frac{1}{a} \tanh^{-1} \frac{u}{a} \quad u^2 < a^2$
- 42 $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \frac{u}{a}$
- 43 $\int \frac{du}{\sqrt{u^2 + a^2}} = \ln(u + \sqrt{u^2 + a^2}) \quad \text{o} \quad \sinh^{-1} \frac{u}{a}$
- 44 $\int \frac{du}{\sqrt{u^2 - a^2}} = \ln(u + \sqrt{u^2 - a^2})$
- 45 $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{u}{a} \right|$
- 46 $\int \frac{du}{u\sqrt{u^2 + a^2}} = -\frac{1}{a} \ln \left(\frac{a + \sqrt{u^2 + a^2}}{u} \right)$
- 47 $\int \frac{du}{u\sqrt{a^2 - u^2}} = -\frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - u^2}}{u} \right)$
- 48 $\int f^{(n)} g \, dx = f^{(n-1)} g - f^{(n-2)} g' + f^{(n-3)} g'' - \dots (-1)^n \int f g^{(n)} \, dx$

METODO DE SUSTITUCION

- | | | |
|----|--|------------------------|
| 49 | $\int F(ax + b) \, dx = \frac{1}{a} \int F(u) \, du$ | $u = ax + b$ |
| 50 | $\int F(\sqrt{ax + b}) \, dx = \frac{2}{a} \int u F(u) \, du$ | $u = \sqrt{ax + b}$ |
| 51 | $\int F(\sqrt[n]{ax + b}) \, dx = \frac{n}{a} \int u^{n-1} F(u) \, du$ | $u = \sqrt[n]{ax + b}$ |
| 52 | $\int F(\sqrt{a^2 - x^2}) \, dx = a \int F(a \cos u) \cos u \, du$ | $x = a \sin u$ |
| 53 | $\int F(\sqrt{x^2 + a^2}) \, dx = a \int F(a \sec u) \sec^2 u \, du$ | $x = a \tan u$ |

$$54 \quad \int F(\sqrt{x^2 - a^2}) dx = a \int F(a \tan u) \sec u \tan u du \quad x = a \sec u$$

$$55 \quad \int F(e^{ax}) dx = \frac{1}{a} \int \frac{F(u)}{u} du \quad u = e^{ax}$$

$$56 \quad \int F(\ln x) dx = \int F(u) e^u du \quad u = \ln x$$

$$57 \quad \int F\left(\sin^{-1} \frac{x}{a}\right) dx = a \int F(u) \cos u du \quad u = \sin^{-1} \frac{x}{a}$$

Para otras funciones trigonométricas recíprocas se obtienen similares resultados.

$$58 \quad \int F(\sin x, \cos x) dx = 2 \int F\left(\frac{2u}{1+u^2}, \frac{1-u^2}{1+u^2}\right) \frac{du}{1+u^2} \quad u = \tan \frac{x}{2}$$

INTEGRALES CLASIFICADAS POR TIPO

INTEGRALES DEL TIPO $ax + b$

$$59 \quad \int \frac{dx}{ax + b} = \frac{1}{a} \ln(ax + b)$$

$$60 \quad \int \frac{x dx}{ax + b} = \frac{x}{a} - \frac{b}{a^2} \ln(ax + b)$$

$$61 \quad \int \frac{x^2 dx}{ax + b} = \frac{(ax + b)^2}{2a^3} - \frac{2b(ax + b)}{a^3} + \frac{b^2}{a^3} \ln(ax + b)$$

$$62 \quad \int \frac{x^3 dx}{ax + b} = \frac{(ax + b)^3}{3a^4} - \frac{3b(ax + b)^2}{2a^4} + \frac{3b^2(ax + b)}{a^4} - \frac{b^3}{a^4} \ln(ax + b)$$

$$63 \quad \int \frac{dx}{x(ax + b)} = \frac{1}{b} \ln\left(\frac{x}{ax + b}\right)$$

$$64 \quad \int \frac{dx}{x^2(ax + b)} = -\frac{1}{bx} + \frac{a}{b^2} \ln\left(\frac{ax + b}{x}\right)$$

$$65 \quad \int \frac{dx}{x^3(ax + b)} = \frac{2ax - b}{2b^2x^2} + \frac{a^2}{b^3} \ln\left(\frac{x}{ax + b}\right)$$

$$66 \quad \int \frac{dx}{(ax + b)^2} = \frac{-1}{a(ax + b)}$$

$$67 \quad \int \frac{x dx}{(ax + b)^2} = \frac{b}{a^2(ax + b)} + \frac{1}{a^2} \ln(ax + b)$$

$$68 \quad \int \frac{x^2 dx}{(ax + b)^2} = \frac{ax + b}{a^3} - \frac{b^2}{a^3(ax + b)} - \frac{2b}{a^3} \ln(ax + b)$$

$$69 \quad \int \frac{x^3 dx}{(ax + b)^2} = \frac{(ax + b)^2}{2a^4} - \frac{3b(ax + b)}{a^4} + \frac{b^3}{a^4(ax + b)} + \frac{3b^2}{a^4} \ln(ax + b)$$

$$70 \quad \int \frac{dx}{x(ax + b)^2} = \frac{1}{b(ax + b)} + \frac{1}{b^2} \ln\left(\frac{x}{ax + b}\right)$$

$$71 \quad \int \frac{dx}{x^2(ax + b)^2} = \frac{-a}{b^2(ax + b)} - \frac{1}{b^2x} + \frac{2a}{b^3} \ln\left(\frac{ax + b}{x}\right)$$

$$72 \quad \int \frac{dx}{x^3(ax+b)^2} = -\frac{(ax+b)^2}{2b^4x^2} + \frac{3a(ax+b)}{b^4x} - \frac{a^3x}{b^4(ax+b)} - \frac{3a^2}{b^4} \ln \left(\frac{ax+b}{x} \right)$$

$$73 \quad \int \frac{dx}{(ax+b)^3} = \frac{-1}{2(ax+b)^2}$$

$$74 \quad \int \frac{x dx}{(ax+b)^3} = \frac{-1}{a^2(ax+b)} + \frac{b}{2a^2(ax+b)^2}$$

$$75 \quad \int \frac{x^2 dx}{(ax+b)^3} = \frac{2b}{a^3(ax+b)} - \frac{b^2}{2a^3(ax+b)^2} + \frac{1}{a^3} \ln(ax+b)$$

$$76 \quad \int \frac{x^3 dx}{(ax+b)^3} = \frac{x}{a^3} - \frac{3b^2}{a^4(ax+b)} + \frac{b^3}{2a^4(ax+b)^2} - \frac{3b}{a^4} \ln(ax+b)$$

$$77 \quad \int \frac{dx}{x(ax+b)^3} = \frac{a^2x^2}{2b^3(ax+b)^2} - \frac{2ax}{b^3(ax+b)} - \frac{1}{b^3} \ln \left(\frac{ax+b}{x} \right)$$

$$78 \quad \int \frac{dx}{x^2(ax+b)^3} = \frac{-a}{2b^2(ax+b)^2} - \frac{2a}{b^3(ax+b)} - \frac{1}{b^3x} + \frac{3a}{b^4} \ln \left(\frac{ax+b}{x} \right)$$

$$79 \quad \int \frac{dx}{x^3(ax+b)^3} = \frac{a^4x^2}{2b^5(ax+b)^2} - \frac{4a^3x}{b^5(ax+b)} - \frac{(ax+b)^2}{2b^5x^2} - \frac{6a^2}{b^5} \ln \left(\frac{ax+b}{x} \right)$$

$$80 \quad \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{(n+1)a} \quad \text{Si } n = -1, \text{ véase } 59.$$

$$81 \quad \int x(ax+b)^n dx = \frac{(ax+b)^{n+2}}{(n+2)a^2} - \frac{b(ax+b)^{n+1}}{(n+1)a^2}, \quad n \neq -1, -2$$

Si $n = -1, -2$, véase 62 67.

$$82 \quad \int x^2(ax+b)^n dx = \frac{(ax+b)^{n+3}}{(n+3)a^3} - \frac{2b(ax+b)^{n+2}}{(n+2)a^3} + \frac{b^2(ax+b)^{n+1}}{(n+1)a^3}$$

Si $n = -1, -2, -3$, véase 61 68 75.

$$83 \quad \int x^m(ax+b)^n dx = \begin{cases} \frac{x^{m+1}(ax+b)^n}{m+n+1} + \frac{nb}{m+n+1} \int x^m(ax+b)^{n-1} dx \\ \frac{x^m(ax+b)^{n+1}}{(m+n+1)a} - \frac{mb}{(m+n+1)a} \int x^{m-1}(ax+b)^n dx \\ \frac{-x^{m+1}(ax+b)^{n+1}}{(n+1)b} + \frac{m+n+2}{(n+1)b} \int x^m(ax+b)^{n+1} dx \end{cases}$$

INTEGRALES DEL TIPO $\sqrt{ax+b}$

$$84 \quad \int \frac{dx}{\sqrt{ax+b}} = \frac{2\sqrt{ax+b}}{a}$$

$$85 \quad \int \frac{x dx}{\sqrt{ax+b}} = \frac{2(ax-2b)}{3a^2} \sqrt{ax+b}$$

$$86 \quad \int \frac{x^2 dx}{\sqrt{ax+b}} = \frac{2(3a^2x^2 - 4abx + 8b^2)}{15a^3} \sqrt{ax+b}$$

$$87 \quad \int \frac{dx}{x\sqrt{ax+b}} = \begin{cases} \frac{1}{\sqrt{b}} \ln \left(\frac{\sqrt{ax+b} - \sqrt{b}}{\sqrt{ax+b} + \sqrt{b}} \right) \\ \frac{2}{\sqrt{-b}} \tan^{-1} \sqrt{\frac{ax+b}{-b}} \end{cases}$$

$$88 \quad \int \frac{dx}{x^2\sqrt{ax+b}} = -\frac{\sqrt{ax+b}}{bx} - \frac{a}{2b} \int \frac{dx}{x\sqrt{ax+b}} \quad \text{Véase } 87$$

- $$\begin{aligned}
 89 \quad \int \sqrt{ax+b} \, dx &= \frac{2\sqrt{(ax+b)^3}}{3a} \\
 90 \quad \int x\sqrt{ax+b} \, dx &= \frac{2(3ax-2b)}{15a^2} \sqrt{(ax+b)^3} \\
 91 \quad \int x^2\sqrt{ax+b} \, dx &= \frac{2(15a^2x^2-12abx+8b^2)}{105a^3} \sqrt{(ax+b)^3} \\
 92 \quad \int \frac{\sqrt{ax+b}}{x} \, dx &= 2\sqrt{ax+b} + b \int \frac{dx}{x\sqrt{ax+b}} \quad \text{Véase } 87 \\
 93 \quad \int \frac{\sqrt{ax+b}}{x^2} \, dx &= -\frac{\sqrt{ax+b}}{x} + \frac{a}{2} \int \frac{dx}{x\sqrt{ax+b}} \quad \text{Véase } 87 \\
 94 \quad \int \frac{x^m}{\sqrt{ax+b}} \, dx &= \frac{2x^m\sqrt{ax+b}}{(2m+1)a} - \frac{2mb}{(2m+1)a} \int \frac{x^{m-1}}{\sqrt{ax+b}} \, dx \\
 95 \quad \int \frac{dx}{x^m\sqrt{ax+b}} &= -\frac{\sqrt{ax+b}}{(m-1)bx^{m-1}} - \frac{(2m-3)a}{(2m-2)b} \int \frac{dx}{x^{m-1}\sqrt{ax+b}} \\
 96 \quad \int x^m\sqrt{ax+b} \, dx &= \frac{2x^m}{(2m+3)a} (ax+b)^{3/2} - \frac{2mb}{(2m+3)a} \int x^{m-1}\sqrt{ax+b} \, dx \\
 97 \quad \int \frac{\sqrt{ax+b}}{x^m} \, dx &= -\frac{\sqrt{ax+b}}{(m-1)x^{m-1}} + \frac{a}{2(m-1)} \int \frac{dx}{x^{m-1}\sqrt{ax+b}} \\
 98 \quad \int \frac{\sqrt{ax+b}}{x^m} \, dx &= \frac{-(ax+b)^{3/2}}{(m-1)bx^{m-1}} - \frac{(2m-5)a}{(2m-2)b} \int \frac{\sqrt{ax+b}}{x^{m-1}} \, dx \\
 99 \quad \int (ax+b)^{m/2} \, dx &= \frac{2(ax+b)^{(m+2)/2}}{a(m+2)} \\
 100 \quad \int x(ax+b)^{m/2} \, dx &= \frac{2(ax+b)^{(m+4)/2}}{a^2(m+4)} - \frac{2b(ax+b)^{(m+2)/2}}{a^2(m+2)} \\
 101 \quad \int x^2(ax+b)^{m/2} \, dx &= \frac{2(ax+b)^{(m+6)/2}}{a^3(m+6)} - \frac{4b(ax+b)^{(m+4)/2}}{a^3(m+4)} + \frac{2b^2(ax+b)^{(m+2)/2}}{a^3(m+2)} \\
 102 \quad \int \frac{(ax+b)^{m/2}}{x} \, dx &= \frac{2(ax+b)^{m/2}}{m} + b \int \frac{(ax+b)^{(m-2)/2}}{x} \, dx \\
 103 \quad \int \frac{(ax+b)^{m/2}}{x^2} \, dx &= -\frac{(ax+b)^{(m+2)/2}}{bx} + \frac{ma}{2b} \int \frac{(ax+b)^{m/2}}{x} \, dx \\
 104 \quad \int \frac{dx}{x(ax+b)^{m/2}} &= \frac{2}{(m-2)b(ax+b)^{(m-2)/2}} + \frac{1}{b} \int \frac{dx}{x(ax+b)^{(m-2)/2}}
 \end{aligned}$$

INTEGRALES DEL TIPO $px+q$ y $ax+b$

- $$\begin{aligned}
 105 \quad \int \frac{dx}{(ax+b)(px+q)} &= \frac{1}{bp-aq} \ln \left(\frac{px+q}{ax+b} \right) \\
 106 \quad \int \frac{x \, dx}{(ax+b)(px+q)} &= \frac{1}{bp-aq} \left\{ \frac{b}{a} \ln(ax+b) - \frac{q}{p} \ln(px+q) \right\} \\
 107 \quad \int \frac{dx}{(ax+b)^2(px+q)} &= \frac{1}{bp-aq} \left\{ \frac{1}{ax+b} + \frac{p}{bp-aq} \ln \left(\frac{px+q}{ax+b} \right) \right\} \\
 108 \quad \int \frac{x \, dx}{(ax+b)^2(px+q)} &= \frac{1}{bp-aq} \left\{ \frac{q}{bp-aq} \ln \left(\frac{ax+b}{px+q} \right) - \frac{b}{a(ax+b)} \right\} \\
 109 \quad \int \frac{x^2 \, dx}{(ax+b)^2(px+q)} &= \frac{b^2}{(bp-aq)a^2(ax+b)} + \frac{1}{(bp-aq)^2} \left\{ \frac{q^2}{p} \ln(px+q) + \frac{b(bp-2aq)}{a^2} \ln(ax+b) \right\}
 \end{aligned}$$

$$110 \quad \int \frac{dx}{(ax+b)^m(px+q)^n} = \frac{-1}{(n-1)(bp-aq)} \left\{ \frac{1}{(ax+b)^{m-1}(px+q)^{n-1}} + a(m+n-2) \int \frac{dx}{(ax+b)^m(px+q)^{n-1}} \right\}$$

$$111 \quad \int \frac{ax+b}{px+q} dx = \frac{ax}{p} + \frac{bp-aq}{p^2} \ln(px+q)$$

$$112 \quad \int \frac{(ax+b)^m}{(px+q)^n} dx = \begin{cases} \frac{-1}{(n-1)(bp-aq)} \left\{ \frac{(ax+b)^{m+1}}{(px+q)^{n-1}} + (n-m-2)a \int \frac{(ax+b)^m}{(px+q)^{n-1}} dx \right\} \\ \frac{-1}{(n-m-1)p} \left\{ \frac{(ax+b)^m}{(px+q)^{n-1}} + m(bp-aq) \int \frac{(ax+b)^{m-1}}{(px+q)^n} dx \right\} \\ \frac{-1}{(n-1)p} \left\{ \frac{(ax+b)^m}{(px+q)^{n-1}} - ma \int \frac{(ax+b)^{m-1}}{(px+q)^{n-1}} dx \right\} \end{cases}$$

INTEGRALES DEL TIPO $px+q$ y $\sqrt{ax+b}$

$$113 \quad \int \frac{px+q}{\sqrt{ax+b}} dx = \frac{2(apx+3aq-2bp)}{3a^2} \sqrt{ax+b}$$

$$114 \quad \int \frac{dx}{(px+q)\sqrt{ax+b}} = \begin{cases} \frac{1}{\sqrt{bp-aq}\sqrt{p}} \ln \left(\frac{\sqrt{p(ax+b)} - \sqrt{bp-aq}}{\sqrt{p(ax+b)} + \sqrt{bp-aq}} \right) \\ \frac{2}{\sqrt{aq-bp}\sqrt{p}} \tan^{-1} \sqrt{\frac{p(ax+b)}{aq-bp}} \end{cases}$$

$$115 \quad \int \frac{\sqrt{ax+b}}{px+q} dx = \begin{cases} \frac{2\sqrt{ax+b}}{p} + \frac{\sqrt{bp-aq}}{p\sqrt{p}} \ln \left(\frac{\sqrt{p(ax+b)} - \sqrt{bp-aq}}{\sqrt{p(ax+b)} + \sqrt{bp-aq}} \right) \\ \frac{2\sqrt{ax+b}}{p} - \frac{2\sqrt{aq-bp}}{p\sqrt{p}} \tan^{-1} \sqrt{\frac{p(ax+b)}{aq-bp}} \end{cases}$$

$$116 \quad \int (px+q)^n \sqrt{ax+b} dx = \frac{2(px+q)^{n+1} \sqrt{ax+b}}{(2n+3)p} + \frac{bp-aq}{(2n+3)p} \int \frac{(px+q)^n}{\sqrt{ax+b}} dx$$

$$117 \quad \int \frac{dx}{(px+q)^n \sqrt{ax+b}} = \frac{\sqrt{ax+b}}{(n-1)(aq-bp)(px+q)^{n-1}} + \frac{(2n-3)a}{2(n-1)(aq-bp)} \int \frac{dx}{(px+q)^{n-1} \sqrt{ax+b}}$$

$$118 \quad \int \frac{(px+q)^n}{\sqrt{ax+b}} dx = \frac{2(px+q)^n \sqrt{ax+b}}{(2n+1)a} + \frac{2n(aq-bp)}{(2n+1)a} \int \frac{(px+q)^{n-1} dx}{\sqrt{ax+b}}$$

$$119 \quad \int \frac{\sqrt{ax+b}}{(px+q)^n} dx = \frac{-\sqrt{ax+b}}{(n-1)p(px+q)^{n-1}} + \frac{a}{2(n-1)p} \int \frac{dx}{(px+q)^{n-1} \sqrt{ax+b}}$$

INTEGRALES DEL TIPO $\sqrt{px+q}$ y $\sqrt{ax+b}$

$$120 \quad \int \frac{dx}{\sqrt{(ax+b)(px+q)}} = \begin{cases} \frac{2}{\sqrt{ap}} \ln(\sqrt{a(px+q)} + \sqrt{p(ax+b)}) \\ \frac{2}{\sqrt{-ap}} \tan^{-1} \sqrt{\frac{-p(ax+b)}{a(px+q)}} \end{cases}$$

$$121 \quad \int \frac{x dx}{\sqrt{(ax+b)(px+q)}} = \frac{\sqrt{(ax+b)(px+q)}}{ap} - \frac{bp+aq}{2ap} \int \frac{dx}{\sqrt{(ax+b)(px+q)}}$$

$$122 \quad \int \sqrt{(ax+b)(px+q)} dx = \frac{2apx + bp + aq}{4ap} \sqrt{(ax+b)(px+q)} - \frac{(bp-aq)^2}{8ap} \int \frac{dx}{\sqrt{(ax+b)(px+q)}}$$

$$123 \quad \int \sqrt{\frac{px+q}{ax+b}} dx = \frac{\sqrt{(ax+b)(px+q)}}{a} + \frac{aq-bp}{2a} \int \frac{dx}{\sqrt{(ax+b)(px+q)}}$$

$$124 \quad \int \frac{dx}{(px+q) \sqrt{(ax+b)(px+q)}} = \frac{2\sqrt{ax+b}}{(aq-bp) \sqrt{px+q}}$$

INTEGRALES DEL TIPO

$$x^2 + a^2$$

$$125 \quad \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$126 \quad \int \frac{x dx}{x^2 + a^2} = \frac{1}{2} \ln(x^2 + a^2)$$

$$127 \quad \int \frac{x^2 dx}{x^2 + a^2} = x - a \tan^{-1} \frac{x}{a}$$

$$128 \quad \int \frac{x^3 dx}{x^2 + a^2} = \frac{x^2}{2} - \frac{a^2}{2} \ln(x^2 + a^2)$$

$$129 \quad \int \frac{dx}{x(x^2 + a^2)} = \frac{1}{2a^2} \ln \left(\frac{x^2}{x^2 + a^2} \right)$$

$$130 \quad \int \frac{dx}{x^2(x^2 + a^2)} = -\frac{1}{a^2 x} - \frac{1}{a^3} \tan^{-1} \frac{x}{a}$$

$$131 \quad \int \frac{dx}{x^3(x^2 + a^2)} = -\frac{1}{2a^2 x^2} - \frac{1}{2a^4} \ln \left(\frac{x^2}{x^2 + a^2} \right)$$

$$132 \quad \int \frac{dx}{(x^2 + a^2)^2} = \frac{x}{2a^2(x^2 + a^2)} + \frac{1}{2a^3} \tan^{-1} \frac{x}{a}$$

$$133 \quad \int \frac{x dx}{(x^2 + a^2)^2} = \frac{-1}{2(x^2 + a^2)}$$

$$134 \quad \int \frac{x^2 dx}{(x^2 + a^2)^2} = \frac{-x}{2(x^2 + a^2)} + \frac{1}{2a} \tan^{-1} \frac{x}{a}$$

$$135 \quad \int \frac{x^3 dx}{(x^2 + a^2)^2} = \frac{a^2}{2(x^2 + a^2)} + \frac{1}{2} \ln(x^2 + a^2)$$

$$136 \quad \int \frac{dx}{x(x^2 + a^2)^2} = \frac{1}{2a^2(x^2 + a^2)} + \frac{1}{2a^4} \ln \left(\frac{x^2}{x^2 + a^2} \right)$$

$$137 \quad \int \frac{dx}{x^2(x^2 + a^2)^2} = -\frac{1}{a^4 x} - \frac{x}{2a^4(x^2 + a^2)} - \frac{3}{2a^5} \tan^{-1} \frac{x}{a}$$

$$138 \quad \int \frac{dx}{x^3(x^2 + a^2)^2} = -\frac{1}{2a^4 x^2} - \frac{1}{2a^4(x^2 + a^2)} - \frac{1}{a^6} \ln \left(\frac{x^2}{x^2 + a^2} \right)$$

$$139 \quad \int \frac{dx}{(x^2 + a^2)^n} = \frac{x}{2(n-1)a^2(x^2 + a^2)^{n-1}} + \frac{2n-3}{(2n-2)a^2} \int \frac{dx}{(x^2 + a^2)^{n-1}}$$

$$140 \quad \int \frac{x dx}{(x^2 + a^2)^n} = \frac{-1}{2(n-1)(x^2 + a^2)^{n-1}}$$

$$141 \quad \int \frac{dx}{x(x^2 + a^2)^n} = \frac{1}{2(n-1)a^2(x^2 + a^2)^{n-1}} + \frac{1}{a^2} \int \frac{dx}{x(x^2 + a^2)^{n-1}}$$

$$142 \quad \int \frac{x^m dx}{(x^2 + a^2)^n} = \int \frac{x^{m-2} dx}{(x^2 + a^2)^{n-1}} - a^2 \int \frac{x^{m-2} dx}{(x^2 + a^2)^n}$$

$$143 \quad \int \frac{dx}{x^m(x^2 + a^2)^n} = \frac{1}{a^2} \int \frac{dx}{x^m(x^2 + a^2)^{n-1}} - \frac{1}{a^2} \int \frac{dx}{x^{m-2}(x^2 + a^2)^n}$$

- $$144 \quad \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left(\frac{x-a}{x+a} \right) \quad \text{o} \quad -\frac{1}{a} \coth^{-1} \frac{x}{a}$$
- $$145 \quad \int \frac{x dx}{x^2 - a^2} = \frac{1}{2} \ln (x^2 - a^2)$$
- $$146 \quad \int \frac{x^2 dx}{x^2 - a^2} = x + \frac{a}{2} \ln \left(\frac{x-a}{x+a} \right)$$
- $$147 \quad \int \frac{x^3 dx}{x^2 - a^2} = \frac{x^2}{2} + \frac{a^2}{2} \ln (x^2 - a^2)$$
- $$148 \quad \int \frac{dx}{x(x^2 - a^2)} = \frac{1}{2a^2} \ln \left(\frac{x^2 - a^2}{x^2} \right)$$
- $$149 \quad \int \frac{dx}{x^2(x^2 - a^2)} = \frac{1}{a^2 x} + \frac{1}{2a^3} \ln \left(\frac{x-a}{x+a} \right)$$
- $$150 \quad \int \frac{dx}{x^3(x^2 - a^2)} = \frac{1}{2a^2 x^2} - \frac{1}{2a^4} \ln \left(\frac{x^2}{x^2 - a^2} \right)$$
- $$151 \quad \int \frac{dx}{(x^2 - a^2)^2} = \frac{-x}{2a^2(x^2 - a^2)} - \frac{1}{4a^3} \ln \left(\frac{x-a}{x+a} \right)$$
- $$152 \quad \int \frac{x dx}{(x^2 - a^2)^2} = \frac{-1}{2(x^2 - a^2)}$$
- $$153 \quad \int \frac{x^2 dx}{(x^2 - a^2)^2} = \frac{-x}{2(x^2 - a^2)} + \frac{1}{4a} \ln \left(\frac{x-a}{x+a} \right)$$
- $$154 \quad \int \frac{x^3 dx}{(x^2 - a^2)^2} = \frac{-a^2}{2(x^2 - a^2)} + \frac{1}{2} \ln (x^2 - a^2)$$
- $$155 \quad \int \frac{dx}{x(x^2 - a^2)^2} = \frac{-1}{2a^2(x^2 - a^2)} + \frac{1}{2a^4} \ln \left(\frac{x^2}{x^2 - a^2} \right)$$
- $$156 \quad \int \frac{dx}{x^2(x^2 - a^2)^2} = -\frac{1}{a^4 x} - \frac{x}{2a^4(x^2 - a^2)} - \frac{3}{4a^5} \ln \left(\frac{x-a}{x+a} \right)$$
- $$157 \quad \int \frac{dx}{x^3(x^2 - a^2)^2} = -\frac{1}{2a^4 x^2} - \frac{1}{2a^4(x^2 - a^2)} + \frac{1}{a^6} \ln \left(\frac{x^2}{x^2 - a^2} \right)$$
- $$158 \quad \int \frac{dx}{(x^2 - a^2)^n} = \frac{-x}{2(n-1)a^2(x^2 - a^2)^{n-1}} - \frac{2n-3}{(2n-2)a^2} \int \frac{dx}{(x^2 - a^2)^{n-1}}$$
- $$159 \quad \int \frac{x dx}{(x^2 - a^2)^n} = \frac{-1}{2(n-1)(x^2 - a^2)^{n-1}}$$
- $$160 \quad \int \frac{dx}{x(x^2 - a^2)^n} = \frac{-1}{2(n-1)a^2(x^2 - a^2)^{n-1}} - \frac{1}{a^2} \int \frac{dx}{x(x^2 - a^2)^{n-1}}$$
- $$161 \quad \int \frac{x^m dx}{(x^2 - a^2)^n} = \int \frac{x^{m-2} dx}{(x^2 - a^2)^{n-1}} + a^2 \int \frac{x^{m-2} dx}{(x^2 - a^2)^n}$$
- $$162 \quad \int \frac{dx}{x^m(x^2 - a^2)^n} = \frac{1}{a^2} \int \frac{dx}{x^{m-2}(x^2 - a^2)^n} - \frac{1}{a^2} \int \frac{dx}{x^m(x^2 - a^2)^{n-1}}$$

$$163 \quad \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left(\frac{a+x}{a-x} \right) \quad \text{o} \quad \frac{1}{a} \tanh^{-1} \frac{x}{a}$$

$$164 \quad \int \frac{x dx}{a^2 - x^2} = -\frac{1}{2} \ln (a^2 - x^2)$$

$$165 \quad \int \frac{x^2 dx}{a^2 - x^2} = -x + \frac{a}{2} \ln \left(\frac{a+x}{a-x} \right)$$

$$166 \quad \int \frac{x^3 dx}{a^2 - x^2} = -\frac{x^2}{2} - \frac{a^2}{2} \ln (a^2 - x^2)$$

$$167 \quad \int \frac{dx}{x(a^2 - x^2)} = \frac{1}{2a^2} \ln \left(\frac{x^2}{a^2 - x^2} \right)$$

$$168 \quad \int \frac{dx}{x^2(a^2 - x^2)} = -\frac{1}{a^2 x} + \frac{1}{2a^3} \ln \left(\frac{a+x}{a-x} \right)$$

$$169 \quad \int \frac{dx}{x^3(a^2 - x^2)} = -\frac{1}{2a^2 x^2} + \frac{1}{2a^4} \ln \left(\frac{x^2}{a^2 - x^2} \right)$$

$$170 \quad \int \frac{dx}{(a^2 - x^2)^2} = \frac{x}{2a^2(a^2 - x^2)} + \frac{1}{4a^3} \ln \left(\frac{a+x}{a-x} \right)$$

$$171 \quad \int \frac{x dx}{(a^2 - x^2)^2} = \frac{1}{2(a^2 - x^2)}$$

$$172 \quad \int \frac{x^2 dx}{(a^2 - x^2)^2} = \frac{x}{2(a^2 - x^2)} - \frac{1}{4a} \ln \left(\frac{a+x}{a-x} \right)$$

$$173 \quad \int \frac{x^3 dx}{(a^2 - x^2)^2} = \frac{a^2}{2(a^2 - x^2)} + \frac{1}{2} \ln (a^2 - x^2)$$

$$174 \quad \int \frac{dx}{x(a^2 - x^2)^2} = \frac{1}{2a^2(a^2 - x^2)} + \frac{1}{2a^4} \ln \left(\frac{x^2}{a^2 - x^2} \right)$$

$$175 \quad \int \frac{dx}{x^2(a^2 - x^2)^2} = \frac{-1}{a^4 x} + \frac{x}{2a^4(a^2 - x^2)} + \frac{3}{4a^5} \ln \left(\frac{a+x}{a-x} \right)$$

$$176 \quad \int \frac{dx}{x^3(a^2 - x^2)^2} = \frac{-1}{2a^4 x^2} + \frac{1}{2a^4(a^2 - x^2)} + \frac{1}{a^6} \ln \left(\frac{x^2}{a^2 - x^2} \right)$$

$$177 \quad \int \frac{dx}{(a^2 - x^2)^n} = \frac{x}{2(n-1)a^2(a^2 - x^2)^{n-1}} + \frac{2n-3}{(2n-2)a^2} \int \frac{dx}{(a^2 - x^2)^{n-1}}$$

$$178 \quad \int \frac{x dx}{(a^2 - x^2)^n} = \frac{1}{2(n-1)(a^2 - x^2)^{n-1}}$$

$$179 \quad \int \frac{dx}{x(a^2 - x^2)^n} = \frac{1}{2(n-1)a^2(a^2 - x^2)^{n-1}} + \frac{1}{a^2} \int \frac{dx}{x(a^2 - x^2)^{n-1}}$$

$$180 \quad \int \frac{x^m dx}{(a^2 - x^2)^n} = a^2 \int \frac{x^{m-2} dx}{(a^2 - x^2)^n} - \int \frac{x^{m-2} dx}{(a^2 - x^2)^{n-1}}$$

$$181 \quad \int \frac{dx}{x^m(a^2 - x^2)^n} = \frac{1}{a^2} \int \frac{dx}{x^{m-2}(a^2 - x^2)^{n-1}} + \frac{1}{a^2} \int \frac{dx}{x^{m-2}(a^2 - x^2)^n}$$

- 182 $\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2}) \quad \text{o} \quad \sinh^{-1} \frac{x}{a}$
- 183 $\int \frac{x dx}{\sqrt{x^2 + a^2}} = \sqrt{x^2 + a^2}$
- 184 $\int \frac{x^2 dx}{\sqrt{x^2 + a^2}} = \frac{x\sqrt{x^2 + a^2}}{2} - \frac{a^2}{2} \ln(x + \sqrt{x^2 + a^2})$
- 185 $\int \frac{x^3 dx}{\sqrt{x^2 + a^2}} = \frac{(x^2 + a^2)^{3/2}}{3} - a^2 \sqrt{x^2 + a^2}$
- 186 $\int \frac{dx}{x\sqrt{x^2 + a^2}} = -\frac{1}{a} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$
- 187 $\int \frac{dx}{x^2 \sqrt{x^2 + a^2}} = -\frac{\sqrt{x^2 + a^2}}{a^2 x}$
- 188 $\int \frac{dx}{x^3 \sqrt{x^2 + a^2}} = -\frac{\sqrt{x^2 + a^2}}{2a^2 x^2} + \frac{1}{2a^3} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$
- 189 $\int \sqrt{x^2 + a^2} dx = \frac{x\sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \ln(x + \sqrt{x^2 + a^2})$
- 190 $\int x\sqrt{x^2 + a^2} dx = \frac{(x^2 + a^2)^{3/2}}{3}$
- 191 $\int x^2 \sqrt{x^2 + a^2} dx = \frac{x(x^2 + a^2)^{3/2}}{4} - \frac{a^2 x \sqrt{x^2 + a^2}}{8} - \frac{a^4}{8} \ln(x + \sqrt{x^2 + a^2})$
- 192 $\int x^3 \sqrt{x^2 + a^2} dx = \frac{(x^2 + a^2)^{5/2}}{5} - \frac{a^2 (x^2 + a^2)^{3/2}}{3}$
- 193 $\int \frac{\sqrt{x^2 + a^2}}{x} dx = \sqrt{x^2 + a^2} - a \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$
- 194 $\int \frac{\sqrt{x^2 + a^2}}{x^2} dx = -\frac{\sqrt{x^2 + a^2}}{x} + \ln(x + \sqrt{x^2 + a^2})$
- 195 $\int \frac{\sqrt{x^2 + a^2}}{x^3} dx = -\frac{\sqrt{x^2 + a^2}}{2x^2} - \frac{1}{2a} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$
- 196 $\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2 + a^2}}$
- 197 $\int \frac{x dx}{(x^2 + a^2)^{3/2}} = \frac{-1}{\sqrt{x^2 + a^2}}$
- 198 $\int \frac{x^2 dx}{(x^2 + a^2)^{3/2}} = \frac{-x}{\sqrt{x^2 + a^2}} + \ln(x + \sqrt{x^2 + a^2})$
- 199 $\int \frac{x^3 dx}{(x^2 + a^2)^{3/2}} = \sqrt{x^2 + a^2} + \frac{a^2}{\sqrt{x^2 + a^2}}$
- 200 $\int \frac{dx}{x(x^2 + a^2)^{3/2}} = \frac{1}{a^2 \sqrt{x^2 + a^2}} - \frac{1}{a^3} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$
- 201 $\int \frac{dx}{x^2 (x^2 + a^2)^{3/2}} = -\frac{\sqrt{x^2 + a^2}}{a^4 x} - \frac{x}{a^4 \sqrt{x^2 + a^2}}$
- 202 $\int \frac{dx}{x^3 (x^2 + a^2)^{3/2}} = \frac{-1}{2a^2 x^2 \sqrt{x^2 + a^2}} - \frac{3}{2a^4 \sqrt{x^2 + a^2}} + \frac{3}{2a^5} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$

$$\begin{aligned}
203 \quad \int (x^2 + a^2)^{3/2} dx &= \frac{x(x^2 + a^2)^{3/2}}{4} + \frac{3a^2 x \sqrt{x^2 + a^2}}{8} + \frac{3}{8} a^4 \ln(x + \sqrt{x^2 + a^2}) \\
204 \quad \int x(x^2 + a^2)^{3/2} dx &= \frac{(x^2 + a^2)^{5/2}}{5} \\
205 \quad \int x^2(x^2 + a^2)^{3/2} dx &= \frac{x(x^2 + a^2)^{5/2}}{6} - \frac{a^2 x(x^2 + a^2)^{3/2}}{24} - \frac{a^4 x \sqrt{x^2 + a^2}}{16} - \frac{a^6}{16} \ln(x + \sqrt{x^2 + a^2}) \\
206 \quad \int x^3(x^2 + a^2)^{3/2} dx &= \frac{(x^2 + a^2)^{7/2}}{7} - \frac{a^2(x^2 + a^2)^{5/2}}{5} \\
207 \quad \int \frac{(x^2 + a^2)^{3/2}}{x} dx &= \frac{(x^2 + a^2)^{3/2}}{3} + a^2 \sqrt{x^2 + a^2} - a^3 \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right) \\
208 \quad \int \frac{(x^2 + a^2)^{3/2}}{x^2} dx &= -\frac{(x^2 + a^2)^{3/2}}{x} + \frac{3x \sqrt{x^2 + a^2}}{2} + \frac{3}{2} a^2 \ln(x + \sqrt{x^2 + a^2}) \\
209 \quad \int \frac{(x^2 + a^2)^{3/2}}{x^3} dx &= -\frac{(x^2 + a^2)^{3/2}}{2x^2} + \frac{3}{2} \sqrt{x^2 + a^2} - \frac{3}{2} a \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)
\end{aligned}$$

INTEGRALES DEL TIPO

$$\sqrt{x^2 - a^2}$$

$$\begin{aligned}
210 \quad \int \frac{dx}{\sqrt{x^2 - a^2}} &= \ln(x + \sqrt{x^2 - a^2}), \quad \int \frac{x dx}{\sqrt{x^2 - a^2}} = \sqrt{x^2 - a^2} \\
211 \quad \int \frac{x^2 dx}{\sqrt{x^2 - a^2}} &= \frac{x\sqrt{x^2 - a^2}}{2} + \frac{a^2}{2} \ln(x + \sqrt{x^2 - a^2}) \\
212 \quad \int \frac{x^3 dx}{\sqrt{x^2 - a^2}} &= \frac{(x^2 - a^2)^{3/2}}{3} + a^2 \sqrt{x^2 - a^2} \\
213 \quad \int \frac{dx}{x\sqrt{x^2 - a^2}} &= \frac{1}{a} \sec^{-1} \left| \frac{x}{a} \right| \\
214 \quad \int \frac{dx}{x^2 \sqrt{x^2 - a^2}} &= \frac{\sqrt{x^2 - a^2}}{a^2 x} \\
215 \quad \int \frac{dx}{x^3 \sqrt{x^2 - a^2}} &= \frac{\sqrt{x^2 - a^2}}{2a^2 x^2} + \frac{1}{2a^3} \sec^{-1} \left| \frac{x}{a} \right| \\
216 \quad \int \sqrt{x^2 - a^2} dx &= \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \ln(x + \sqrt{x^2 - a^2}) \\
217 \quad \int x\sqrt{x^2 - a^2} dx &= \frac{(x^2 - a^2)^{3/2}}{3} \\
218 \quad \int x^2 \sqrt{x^2 - a^2} dx &= \frac{x(x^2 - a^2)^{3/2}}{4} + \frac{a^2 x \sqrt{x^2 - a^2}}{8} - \frac{a^4}{8} \ln(x + \sqrt{x^2 - a^2}) \\
219 \quad \int x^3 \sqrt{x^2 - a^2} dx &= \frac{(x^2 - a^2)^{5/2}}{5} + \frac{a^2(x^2 - a^2)^{3/2}}{3} \\
220 \quad \int \frac{\sqrt{x^2 - a^2}}{x} dx &= \sqrt{x^2 - a^2} - a \sec^{-1} \left| \frac{x}{a} \right| \\
221 \quad \int \frac{\sqrt{x^2 - a^2}}{x^2} dx &= -\frac{\sqrt{x^2 - a^2}}{x} + \ln(x + \sqrt{x^2 - a^2}) \\
222 \quad \int \frac{\sqrt{x^2 - a^2}}{x^3} dx &= -\frac{\sqrt{x^2 - a^2}}{2x^2} + \frac{1}{2a} \sec^{-1} \left| \frac{x}{a} \right| \\
223 \quad \int \frac{dx}{(x^2 - a^2)^{3/2}} &= -\frac{x}{a^2 \sqrt{x^2 - a^2}}
\end{aligned}$$

- 224 $\int \frac{x dx}{(x^2 - a^2)^{3/2}} = \frac{-1}{\sqrt{x^2 - a^2}}$
- 225 $\int \frac{x^2 dx}{(x^2 - a^2)^{3/2}} = -\frac{x}{\sqrt{x^2 - a^2}} + \ln(x + \sqrt{x^2 - a^2})$
- 226 $\int \frac{x^3 dx}{(x^2 - a^2)^{3/2}} = \sqrt{x^2 - a^2} - \frac{a^2}{\sqrt{x^2 - a^2}}$
- 227 $\int \frac{dx}{x(x^2 - a^2)^{3/2}} = \frac{-1}{a^2 \sqrt{x^2 - a^2}} - \frac{1}{a^3} \sec^{-1} \left| \frac{x}{a} \right|$
- 228 $\int \frac{dx}{x^2(x^2 - a^2)^{3/2}} = -\frac{\sqrt{x^2 - a^2}}{a^4 x} - \frac{x}{a^4 \sqrt{x^2 - a^2}}$
- 229 $\int \frac{dx}{x^3(x^2 - a^2)^{3/2}} = \frac{1}{2a^2 x^2 \sqrt{x^2 - a^2}} - \frac{3}{2a^4 \sqrt{x^2 - a^2}} - \frac{3}{2a^5} \sec^{-1} \left| \frac{x}{a} \right|$
- 230 $\int (x^2 - a^2)^{3/2} dx = \frac{x(x^2 - a^2)^{3/2}}{4} - \frac{3a^2 x \sqrt{x^2 - a^2}}{8} + \frac{3}{8} a^4 \ln(x + \sqrt{x^2 - a^2})$
- 231 $\int x(x^2 - a^2)^{3/2} dx = \frac{(x^2 - a^2)^{5/2}}{5}$
- 232 $\int x^2(x^2 - a^2)^{3/2} dx = \frac{x(x^2 - a^2)^{5/2}}{6} + \frac{a^2 x(x^2 - a^2)^{3/2}}{24} - \frac{a^4 x \sqrt{x^2 - a^2}}{16} + \frac{a^6}{16} \ln(x + \sqrt{x^2 - a^2})$
- 233 $\int x^3(x^2 - a^2)^{3/2} dx = \frac{(x^2 - a^2)^{7/2}}{7} + \frac{a^2(x^2 - a^2)^{5/2}}{5}$
- 234 $\int \frac{(x^2 - a^2)^{3/2}}{x} dx = \frac{(x^2 - a^2)^{3/2}}{3} - a^2 \sqrt{x^2 - a^2} + a^3 \sec^{-1} \left| \frac{x}{a} \right|$
- 235 $\int \frac{(x^2 - a^2)^{3/2}}{x^2} dx = -\frac{(x^2 - a^2)^{3/2}}{x} + \frac{3x \sqrt{x^2 - a^2}}{2} - \frac{3}{2} a^2 \ln(x + \sqrt{x^2 - a^2})$
- 236 $\int \frac{(x^2 - a^2)^{3/2}}{x^3} dx = -\frac{(x^2 - a^2)^{3/2}}{2x^2} + \frac{3\sqrt{x^2 - a^2}}{2} - \frac{3}{2} a \sec^{-1} \left| \frac{x}{a} \right|$

INTEGRALES DEL TIPO

$$\sqrt{a^2 - x^2}$$

- 237 $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a}$
- 238 $\int \frac{x dx}{\sqrt{a^2 - x^2}} = -\sqrt{a^2 - x^2}$
- 239 $\int \frac{x^2 dx}{\sqrt{a^2 - x^2}} = -\frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$
- 240 $\int \frac{x^3 dx}{\sqrt{a^2 - x^2}} = \frac{(a^2 - x^2)^{3/2}}{3} - a^2 \sqrt{a^2 - x^2}$
- 241 $\int \frac{dx}{x\sqrt{a^2 - x^2}} = -\frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$
- 242 $\int \frac{dx}{x^2 \sqrt{a^2 - x^2}} = -\frac{\sqrt{a^2 - x^2}}{a^2 x}$
- 243 $\int \frac{dx}{x^3 \sqrt{a^2 - x^2}} = -\frac{\sqrt{a^2 - x^2}}{2a^2 x^2} - \frac{1}{2a^3} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$

$$244 \quad \int \sqrt{a^2 - x^2} dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \operatorname{sen}^{-1} \frac{x}{a}$$

$$245 \quad \int x\sqrt{a^2 - x^2} dx = -\frac{(a^2 - x^2)^{3/2}}{3}$$

$$246 \quad \int x^2\sqrt{a^2 - x^2} dx = -\frac{x(a^2 - x^2)^{3/2}}{4} + \frac{a^2x\sqrt{a^2 - x^2}}{8} + \frac{a^4}{8} \operatorname{sen}^{-1} \frac{x}{a}$$

$$247 \quad \int x^3\sqrt{a^2 - x^2} dx = \frac{(a^2 - x^2)^{5/2}}{5} - \frac{a^2(a^2 - x^2)^{3/2}}{3}$$

$$248 \quad \int \frac{\sqrt{a^2 - x^2}}{x} dx = \sqrt{a^2 - x^2} - a \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$249 \quad \int \frac{\sqrt{a^2 - x^2}}{x^2} dx = -\frac{\sqrt{a^2 - x^2}}{x} - \operatorname{sen}^{-1} \frac{x}{a}$$

$$250 \quad \int \frac{\sqrt{a^2 - x^2}}{x^3} dx = -\frac{\sqrt{a^2 - x^2}}{2x^2} + \frac{1}{2a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$251 \quad \int \frac{dx}{(a^2 - x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2 - x^2}}$$

$$252 \quad \int \frac{x dx}{(a^2 - x^2)^{3/2}} = \frac{1}{\sqrt{a^2 - x^2}}$$

$$253 \quad \int \frac{x^2 dx}{(a^2 - x^2)^{3/2}} = \frac{x}{\sqrt{a^2 - x^2}} - \operatorname{sen}^{-1} \frac{x}{a}$$

$$254 \quad \int \frac{x^3 dx}{(a^2 - x^2)^{3/2}} = \sqrt{a^2 - x^2} + \frac{a^2}{\sqrt{a^2 - x^2}}$$

$$255 \quad \int \frac{dx}{x(a^2 - x^2)^{3/2}} = \frac{1}{a^2\sqrt{a^2 - x^2}} - \frac{1}{a^3} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$256 \quad \int \frac{dx}{x^2(a^2 - x^2)^{3/2}} = -\frac{\sqrt{a^2 - x^2}}{a^4x} + \frac{x}{a^4\sqrt{a^2 - x^2}}$$

$$257 \quad \int \frac{dx}{x^3(a^2 - x^2)^{3/2}} = \frac{-1}{2a^2x^2\sqrt{a^2 - x^2}} + \frac{3}{2a^4\sqrt{a^2 - x^2}} - \frac{3}{2a^5} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$258 \quad \int (a^2 - x^2)^{3/2} dx = \frac{x(a^2 - x^2)^{3/2}}{4} + \frac{3a^2x\sqrt{a^2 - x^2}}{8} + \frac{3}{8}a^4 \operatorname{sen}^{-1} \frac{x}{a}$$

$$259 \quad \int x(a^2 - x^2)^{3/2} dx = -\frac{(a^2 - x^2)^{5/2}}{5}$$

$$260 \quad \int x^2(a^2 - x^2)^{3/2} dx = -\frac{x(a^2 - x^2)^{5/2}}{6} + \frac{a^2x(a^2 - x^2)^{3/2}}{24} + \frac{a^4x\sqrt{a^2 - x^2}}{16} + \frac{a^6}{16} \operatorname{sen}^{-1} \frac{x}{a}$$

$$261 \quad \int x^3(a^2 - x^2)^{3/2} dx = \frac{(a^2 - x^2)^{7/2}}{7} - \frac{a^2(a^2 - x^2)^{5/2}}{5}$$

$$262 \quad \int \frac{(a^2 - x^2)^{3/2}}{x} dx = \frac{(a^2 - x^2)^{3/2}}{3} + a^2\sqrt{a^2 - x^2} - a^3 \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$263 \quad \int \frac{(a^2 - x^2)^{3/2}}{x^2} dx = -\frac{(a^2 - x^2)^{3/2}}{x} - \frac{3x\sqrt{a^2 - x^2}}{2} - \frac{3}{2}a^2 \operatorname{sen}^{-1} \frac{x}{a}$$

$$264 \quad \int \frac{(a^2 - x^2)^{3/2}}{x^3} dx = -\frac{(a^2 - x^2)^{3/2}}{2x^2} - \frac{3\sqrt{a^2 - x^2}}{2} + \frac{3}{2}a \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$265 \quad \int \frac{dx}{ax^2 + bx + c} = \begin{cases} \frac{2}{\sqrt{4ac - b^2}} \tan^{-1} \frac{2ax + b}{\sqrt{4ac - b^2}} \\ \frac{1}{\sqrt{b^2 - 4ac}} \ln \left(\frac{2ax + b - \sqrt{b^2 - 4ac}}{2ax + b + \sqrt{b^2 - 4ac}} \right) \end{cases}$$

Si $b^2 = 4ac$, se puede escribir $ax^2 + bx + c = a(x + b/2a)^2$ y se emplean los resultados de las páginas 4 - 5.

$$266 \quad \int \frac{x dx}{ax^2 + bx + c} = \frac{1}{2a} \ln(ax^2 + bx + c) - \frac{b}{2a} \int \frac{dx}{ax^2 + bx + c}$$

$$267 \quad \int \frac{x^2 dx}{ax^2 + bx + c} = \frac{x}{a} - \frac{b}{2a^2} \ln(ax^2 + bx + c) + \frac{b^2 - 2ac}{2a^2} \int \frac{dx}{ax^2 + bx + c}$$

$$268 \quad \int \frac{x^m dx}{ax^2 + bx + c} = \frac{x^{m-1}}{(m-1)a} - \frac{c}{a} \int \frac{x^{m-2} dx}{ax^2 + bx + c} - \frac{b}{a} \int \frac{x^{m-1} dx}{ax^2 + bx + c}$$

$$269 \quad \int \frac{dx}{x(ax^2 + bx + c)} = \frac{1}{2c} \ln \left(\frac{x^2}{ax^2 + bx + c} \right) - \frac{b}{2c} \int \frac{dx}{ax^2 + bx + c}$$

$$270 \quad \int \frac{dx}{x^2(ax^2 + bx + c)} = \frac{b}{2c^2} \ln \left(\frac{ax^2 + bx + c}{x^2} \right) - \frac{1}{cx} + \frac{b^2 - 2ac}{2c^2} \int \frac{dx}{ax^2 + bx + c}$$

$$271 \quad \int \frac{dx}{x^n(ax^2 + bx + c)} = -\frac{1}{(n-1)cx^{n-1}} - \frac{b}{c} \int \frac{dx}{x^{n-1}(ax^2 + bx + c)} - \frac{a}{c} \int \frac{dx}{x^{n-2}(ax^2 + bx + c)}$$

$$272 \quad \int \frac{dx}{(ax^2 + bx + c)^2} = \frac{2ax + b}{(4ac - b^2)(ax^2 + bx + c)} + \frac{2a}{4ac - b^2} \int \frac{dx}{ax^2 + bx + c}$$

$$273 \quad \int \frac{x dx}{(ax^2 + bx + c)^2} = -\frac{bx + 2c}{(4ac - b^2)(ax^2 + bx + c)} - \frac{b}{4ac - b^2} \int \frac{dx}{ax^2 + bx + c}$$

$$274 \quad \int \frac{x^2 dx}{(ax^2 + bx + c)^2} = \frac{(b^2 - 2ac)x + bc}{a(4ac - b^2)(ax^2 + bx + c)} + \frac{2c}{4ac - b^2} \int \frac{dx}{ax^2 + bx + c}$$

$$275 \quad \int \frac{x^m dx}{(ax^2 + bx + c)^n} = -\frac{x^{m-1}}{(2n-m-1)a(ax^2 + bx + c)^{n-1}} + \frac{(m-1)c}{(2n-m-1)a} \int \frac{x^{m-2} dx}{(ax^2 + bx + c)^n} \\ - \frac{(n-m)b}{(2n-m-1)a} \int \frac{x^{m-1} dx}{(ax^2 + bx + c)^n}$$

$$276 \quad \int \frac{x^{2n-1} dx}{(ax^2 + bx + c)^n} = \frac{1}{a} \int \frac{x^{2n-3} dx}{(ax^2 + bx + c)^{n-1}} - \frac{c}{a} \int \frac{x^{2n-3} dx}{(ax^2 + bx + c)^n} - \frac{b}{a} \int \frac{x^{2n-2} dx}{(ax^2 + bx + c)^n}$$

$$277 \quad \int \frac{dx}{x(ax^2 + bx + c)^2} = \frac{1}{2c(ax^2 + bx + c)} - \frac{b}{2c} \int \frac{dx}{(ax^2 + bx + c)^2} + \frac{1}{c} \int \frac{dx}{x(ax^2 + bx + c)}$$

$$278 \quad \int \frac{dx}{x^2(ax^2 + bx + c)^2} = -\frac{1}{cx(ax^2 + bx + c)} - \frac{3a}{c} \int \frac{dx}{(ax^2 + bx + c)^2} - \frac{2b}{c} \int \frac{dx}{x(ax^2 + bx + c)^2}$$

$$279 \quad \int \frac{dx}{x^m(ax^2 + bx + c)^n} = -\frac{1}{(m-1)cx^{m-1}(ax^2 + bx + c)^{n-1}} - \frac{(m+2n-3)a}{(m-1)c} \int \frac{dx}{x^{m-2}(ax^2 + bx + c)^n} \\ - \frac{(m+n-2)b}{(m-1)c} \int \frac{dx}{x^{m-1}(ax^2 + bx + c)^n}$$

Si en las fórmulas siguientes $b^2=4ac$ se puede escribir $ax^2+bx+c=\sqrt{a}(x+b/2a)$ y se pueden emplear las fórmulas de las páginas 4 - 5.

$$280 \quad \int \frac{dx}{\sqrt{ax^2 + bx + c}} = \begin{cases} \frac{1}{\sqrt{a}} \ln (2\sqrt{a} \sqrt{ax^2 + bx + c} + 2ax + b) \\ -\frac{1}{\sqrt{-a}} \operatorname{sen}^{-1} \left(\frac{2ax + b}{\sqrt{b^2 - 4ac}} \right) \text{ o } \frac{1}{\sqrt{a}} \operatorname{senh}^{-1} \left(\frac{2ax + b}{\sqrt{4ac - b^2}} \right) \end{cases}$$

$$281 \quad \int \frac{x dx}{\sqrt{ax^2 + bx + c}} = \frac{\sqrt{ax^2 + bx + c}}{a} - \frac{b}{2a} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$282 \quad \int \frac{x^2 dx}{\sqrt{ax^2 + bx + c}} = \frac{2ax - 3b}{4a^2} \sqrt{ax^2 + bx + c} + \frac{3b^2 - 4ac}{8a^2} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$283 \quad \int \frac{dx}{x\sqrt{ax^2 + bx + c}} = \begin{cases} -\frac{1}{\sqrt{c}} \ln \left(\frac{2\sqrt{c} \sqrt{ax^2 + bx + c} + bx + 2c}{x} \right) \\ \frac{1}{\sqrt{-c}} \operatorname{sen}^{-1} \left(\frac{bx + 2c}{|x| \sqrt{b^2 - 4ac}} \right) \text{ o } -\frac{1}{\sqrt{c}} \operatorname{senh}^{-1} \left(\frac{bx + 2c}{|x| \sqrt{4ac - b^2}} \right) \end{cases}$$

$$284 \quad \int \frac{dx}{x^2 \sqrt{ax^2 + bx + c}} = -\frac{\sqrt{ax^2 + bx + c}}{cx} - \frac{b}{2c} \int \frac{dx}{x\sqrt{ax^2 + bx + c}}$$

$$285 \quad \int \sqrt{ax^2 + bx + c} dx = \frac{(2ax + b) \sqrt{ax^2 + bx + c}}{4a} + \frac{4ac - b^2}{8a} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$286 \quad \int x \sqrt{ax^2 + bx + c} dx = \frac{(ax^2 + bx + c)^{3/2}}{3a} - \frac{b(2ax + b)}{8a^2} \sqrt{ax^2 + bx + c} - \frac{b(4ac - b^2)}{16a^2} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$287 \quad \int x^2 \sqrt{ax^2 + bx + c} dx = \frac{6ax - 5b}{24a^2} (ax^2 + bx + c)^{3/2} + \frac{5b^2 - 4ac}{16a^2} \int \sqrt{ax^2 + bx + c} dx$$

$$288 \quad \int \frac{\sqrt{ax^2 + bx + c}}{x} dx = \sqrt{ax^2 + bx + c} + \frac{b}{2} \int \frac{dx}{\sqrt{ax^2 + bx + c}} + c \int \frac{dx}{x\sqrt{ax^2 + bx + c}}$$

$$289 \quad \int \frac{\sqrt{ax^2 + bx + c}}{x^2} dx = -\frac{\sqrt{ax^2 + bx + c}}{x} + a \int \frac{dx}{\sqrt{ax^2 + bx + c}} + \frac{b}{2} \int \frac{dx}{x\sqrt{ax^2 + bx + c}}$$

$$290 \quad \int \frac{dx}{(ax^2 + bx + c)^{3/2}} = \frac{2(2ax + b)}{(4ac - b^2) \sqrt{ax^2 + bx + c}}$$

$$291 \quad \int \frac{x dx}{(ax^2 + bx + c)^{3/2}} = \frac{2(bx + 2c)}{(b^2 - 4ac) \sqrt{ax^2 + bx + c}}$$

$$292 \quad \int \frac{x^2 dx}{(ax^2 + bx + c)^{3/2}} = \frac{(2b^2 - 4ac)x + 2bc}{a(4ac - b^2) \sqrt{ax^2 + bx + c}} + \frac{1}{a} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$293 \quad \int \frac{dx}{x(ax^2 + bx + c)^{3/2}} = \frac{1}{c\sqrt{ax^2 + bx + c}} + \frac{1}{c} \int \frac{dx}{x\sqrt{ax^2 + bx + c}} - \frac{b}{2c} \int \frac{dx}{(ax^2 + bx + c)^{3/2}}$$

$$294 \quad \int \frac{dx}{x^2(ax^2 + bx + c)^{3/2}} = -\frac{ax^2 + 2bx + c}{c^2 x \sqrt{ax^2 + bx + c}} + \frac{b^2 - 2ac}{2c^2} \int \frac{dx}{(ax^2 + bx + c)^{3/2}} - \frac{3b}{2c^2} \int \frac{dx}{x\sqrt{ax^2 + bx + c}}$$

$$295 \quad \int (ax^2 + bx + c)^{n+1/2} dx = \frac{(2ax + b)(ax^2 + bx + c)^{n+1/2}}{4a(n+1)} + \frac{(2n+1)(4ac - b^2)}{8a(n+1)} \int (ax^2 + bx + c)^{n-1/2} dx$$

$$296 \quad \int x(ax^2 + bx + c)^{n+1/2} dx = \frac{(ax^2 + bx + c)^{n+3/2}}{a(2n+3)} - \frac{b}{2a} \int (ax^2 + bx + c)^{n+1/2} dx$$

$$297 \quad \int \frac{dx}{(ax^2 + bx + c)^{n+1/2}} = \frac{2(2ax + b)}{(2n-1)(4ac - b^2)(ax^2 + bx + c)^{n-1/2}} \\ + \frac{8a(n-1)}{(2n-1)(4ac - b^2)} \int \frac{dx}{(ax^2 + bx + c)^{n-1/2}}$$

$$298 \quad \int \frac{dx}{x(ax^2 + bx + c)^{n+1/2}} = \frac{1}{(2n-1)c(ax^2 + bx + c)^{n-1/2}} \\ + \frac{1}{c} \int \frac{dx}{x(ax^2 + bx + c)^{n-1/2}} - \frac{b}{2c} \int \frac{dx}{(ax^2 + bx + c)^{n+1/2}}$$

INTEGRALES DEL TIPO $x^3 \pm a^3$

para integrales del tipo $x^3 - a^3$ se reemplaza a por $-a$

$$299 \quad \int \frac{dx}{x^3 + a^3} = \frac{1}{6a^2} \ln \frac{(x+a)^2}{x^2 - ax + a^2} + \frac{1}{a^2\sqrt{3}} \tan^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$300 \quad \int \frac{x dx}{x^3 + a^3} = \frac{1}{6a} \ln \frac{x^2 - ax + a^2}{(x+a)^2} + \frac{1}{a\sqrt{3}} \tan^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$301 \quad \int \frac{x^2 dx}{x^3 + a^3} = \frac{1}{3} \ln(x^3 + a^3) \quad 302 \quad \int \frac{dx}{x(x^3 + a^3)} = \frac{1}{3a^3} \ln \left(\frac{x^3}{x^3 + a^3} \right)$$

$$303 \quad \int \frac{dx}{x^2(x^3 + a^3)} = -\frac{1}{a^3x} - \frac{1}{6a^4} \ln \frac{x^2 - ax + a^2}{(x+a)^2} - \frac{1}{a^4\sqrt{3}} \tan^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$304 \quad \int \frac{dx}{(x^3 + a^3)^2} = \frac{x}{3a^3(x^3 + a^3)} + \frac{1}{9a^5} \ln \frac{(x+a)^2}{x^2 - ax + a^2} + \frac{2}{3a^5\sqrt{3}} \tan^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$305 \quad \int \frac{x dx}{(x^3 + a^3)^2} = \frac{x^2}{3a^3(x^3 + a^3)} + \frac{1}{18a^4} \ln \frac{x^2 - ax + a^2}{(x+a)^2} + \frac{1}{3a^4\sqrt{3}} \tan^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$306 \quad \int \frac{x^2 dx}{(x^3 + a^3)^2} = -\frac{1}{3(x^3 + a^3)}$$

$$307 \quad \int \frac{dx}{x(x^3 + a^3)^2} = \frac{1}{3a^3(x^3 + a^3)} + \frac{1}{3a^6} \ln \left(\frac{x^3}{x^3 + a^3} \right)$$

$$308 \quad \int \frac{dx}{x^2(x^3 + a^3)^2} = -\frac{1}{a^6x} - \frac{x^2}{3a^6(x^3 + a^3)} - \frac{4}{3a^6} \int \frac{x dx}{x^3 + a^3} \quad \text{Véase } 300$$

$$309 \quad \int \frac{x^m dx}{x^3 + a^3} = \frac{x^{m-2}}{m-2} - a^3 \int \frac{x^{m-3} dx}{x^3 + a^3}$$

$$310 \quad \int \frac{dx}{x^n(x^3 + a^3)} = \frac{-1}{a^3(n-1)x^{n-1}} - \frac{1}{a^3} \int \frac{dx}{x^{n-3}(x^3 + a^3)}$$

INTEGRALES DEL TIPO $x^4 \pm a^4$

$$311 \quad \int \frac{dx}{x^4 + a^4} = \frac{1}{4a^3\sqrt{2}} \ln \left(\frac{x^2 + ax\sqrt{2} + a^2}{x^2 - ax\sqrt{2} + a^2} \right) - \frac{1}{2a^3\sqrt{2}} \tan^{-1} \frac{ax\sqrt{2}}{x^2 - a^2}$$

$$312 \quad \int \frac{x dx}{x^4 + a^4} = \frac{1}{2a^2} \tan^{-1} \frac{x^2}{a^2}$$

$$313 \quad \int \frac{x^2 dx}{x^4 + a^4} = \frac{1}{4a\sqrt{2}} \ln \left(\frac{x^2 - ax\sqrt{2} + a^2}{x^2 + ax\sqrt{2} + a^2} \right) - \frac{1}{2a\sqrt{2}} \tan^{-1} \frac{ax\sqrt{2}}{x^2 - a^2}$$

$$314 \quad \int \frac{x^3 dx}{x^4 + a^4} = \frac{1}{4} \ln(x^4 + a^4)$$

- $$\begin{aligned}
315 \quad \int \frac{dx}{x(x^4 + a^4)} &= \frac{1}{4a^4} \ln \left(\frac{x^4}{x^4 + a^4} \right) \\
316 \quad \int \frac{dx}{x^2(x^4 + a^4)} &= -\frac{1}{a^4 x} - \frac{1}{4a^5 \sqrt{2}} \ln \left(\frac{x^2 - ax\sqrt{2} + a^2}{x^2 + ax\sqrt{2} + a^2} \right) + \frac{1}{2a^5 \sqrt{2}} \tan^{-1} \frac{ax\sqrt{2}}{x^2 - a^2} \\
317 \quad \int \frac{dx}{x^3(x^4 + a^4)} &= -\frac{1}{2a^4 x^2} - \frac{1}{2a^6} \tan^{-1} \frac{x^2}{a^2} \\
318 \quad \int \frac{dx}{x^4 - a^4} &= \frac{1}{4a^3} \ln \left(\frac{x-a}{x+a} \right) - \frac{1}{2a^3} \tan^{-1} \frac{x}{a} \\
319 \quad \int \frac{x dx}{x^4 - a^4} &= \frac{1}{4a^2} \ln \left(\frac{x^2 - a^2}{x^2 + a^2} \right) \\
320 \quad \int \frac{x^2 dx}{x^4 - a^4} &= \frac{1}{4a} \ln \left(\frac{x-a}{x+a} \right) + \frac{1}{2a} \tan^{-1} \frac{x}{a} \\
321 \quad \int \frac{x^3 dx}{x^4 - a^4} &= \frac{1}{4} \ln (x^4 - a^4) \\
322 \quad \int \frac{dx}{x(x^4 - a^4)} &= \frac{1}{4a^4} \ln \left(\frac{x^4 - a^4}{x^4} \right) \\
323 \quad \int \frac{dx}{x^2(x^4 - a^4)} &= \frac{1}{a^4 x} + \frac{1}{4a^5} \ln \left(\frac{x-a}{x+a} \right) + \frac{1}{2a^5} \tan^{-1} \frac{x}{a} \\
324 \quad \int \frac{dx}{x^3(x^4 - a^4)} &= \frac{1}{2a^4 x^2} + \frac{1}{4a^6} \ln \left(\frac{x^2 - a^2}{x^2 + a^2} \right)
\end{aligned}$$

INTEGRALES DEL TIPO $x^n \pm a^n$

- $$\begin{aligned}
325 \quad \int \frac{dx}{x(x^n + a^n)} &= \frac{1}{na^n} \ln \frac{x^n}{x^n + a^n} \\
326 \quad \int \frac{x^{n-1} dx}{x^n + a^n} &= \frac{1}{n} \ln (x^n + a^n) \\
327 \quad \int \frac{x^m dx}{(x^n + a^n)^r} &= \int \frac{x^{m-n} dx}{(x^n + a^n)^{r-1}} - a^n \int \frac{x^{m-n} dx}{(x^n + a^n)^r} \\
328 \quad \int \frac{dx}{x^m(x^n + a^n)^r} &= \frac{1}{a^n} \int \frac{dx}{x^m(x^n + a^n)^{r-1}} - \frac{1}{a^n} \int \frac{dx}{x^{m-n}(x^n + a^n)^r} \\
329 \quad \int \frac{dx}{x\sqrt{x^n + a^n}} &= \frac{1}{n\sqrt{a^n}} \ln \left(\frac{\sqrt{x^n + a^n} - \sqrt{a^n}}{\sqrt{x^n + a^n} + \sqrt{a^n}} \right) \\
330 \quad \int \frac{dx}{x(x^n - a^n)} &= \frac{1}{na^n} \ln \left(\frac{x^n - a^n}{x^n} \right) \\
331 \quad \int \frac{x^{n-1} dx}{x^n - a^n} &= \frac{1}{n} \ln (x^n - a^n) \\
332 \quad \int \frac{x^m dx}{(x^n - a^n)^r} &= a^n \int \frac{x^{m-n} dx}{(x^n - a^n)^r} + \int \frac{x^{m-n} dx}{(x^n - a^n)^{r-1}} \\
333 \quad \int \frac{dx}{x^m(x^n - a^n)^r} &= \frac{1}{a^n} \int \frac{dx}{x^m(x^n - a^n)^{r-1}} - \frac{1}{a^n} \int \frac{dx}{x^{m-n}(x^n - a^n)^r} \\
334 \quad \int \frac{dx}{x\sqrt{x^n - a^n}} &= \frac{2}{n\sqrt{a^n}} \cos^{-1} \sqrt{\frac{a^n}{x^n}}
\end{aligned}$$

$$335 \quad \int \frac{x^{p-1} dx}{x^{2m} + a^{2m}} = \frac{1}{ma^{2m-p}} \sum_{k=1}^m \operatorname{sen} \frac{(2k-1)p\pi}{2m} \tan^{-1} \left(\frac{x + a \cos [(2k-1)\pi/2m]}{a \operatorname{sen} [(2k-1)\pi/2m]} \right) \\ - \frac{1}{2ma^{2m-p}} \sum_{k=1}^m \cos \frac{(2k-1)p\pi}{2m} \ln \left(x^2 + 2ax \cos \frac{(2k-1)\pi}{2m} + a^2 \right)$$

donde $0 < p \leq 2m$.

$$336 \quad \int \frac{x^{p-1} dx}{x^{2m} - a^{2m}} = \frac{1}{2ma^{2m-p}} \sum_{k=1}^{m-1} \cos \frac{kp\pi}{m} \ln \left(x^2 - 2ax \cos \frac{k\pi}{m} + a^2 \right) \\ - \frac{1}{ma^{2m-p}} \sum_{k=1}^{m-1} \operatorname{sen} \frac{kp\pi}{m} \tan^{-1} \left(\frac{x - a \cos (k\pi/m)}{a \operatorname{sen} (k\pi/m)} \right) \\ + \frac{1}{2ma^{2m-p}} \{ \ln (x-a) + (-1)^p \ln (x+a) \}$$

donde $0 < p \leq 2m$.

$$337 \quad \int \frac{x^{p-1} dx}{x^{2m+1} + a^{2m+1}} = \frac{2(-1)^{p-1}}{(2m+1)a^{2m-p+1}} \sum_{k=1}^m \operatorname{sen} \frac{2kp\pi}{2m+1} \tan^{-1} \left(\frac{x + a \cos [2k\pi/(2m+1)]}{a \operatorname{sen} [2k\pi/(2m+1)]} \right) \\ - \frac{(-1)^{p-1}}{(2m+1)a^{2m-p+1}} \sum_{k=1}^m \cos \frac{2kp\pi}{2m+1} \ln \left(x^2 + 2ax \cos \frac{2k\pi}{2m+1} + a^2 \right) \\ + \frac{(-1)^{p-1} \ln (x+a)}{(2m+1)a^{2m-p+1}}$$

donde $0 < p \leq 2m+1$.

$$338 \quad \int \frac{x^{p-1} dx}{x^{2m+1} - a^{2m+1}} = \frac{-2}{(2m+1)a^{2m-p+1}} \sum_{k=1}^m \operatorname{sen} \frac{2kp\pi}{2m+1} \tan^{-1} \left(\frac{x - a \cos [2k\pi/(2m+1)]}{a \operatorname{sen} [2k\pi/(2m+1)]} \right) \\ + \frac{1}{(2m+1)a^{2m-p+1}} \sum_{k=1}^m \cos \frac{2kp\pi}{2m+1} \ln \left(x^2 - 2ax \cos \frac{2k\pi}{2m+1} + a^2 \right) \\ + \frac{\ln (x-a)}{(2m+1)a^{2m-p+1}}$$

donde $0 < p \leq 2m+1$.

INTEGRALES DEL TIPO $\operatorname{sen} ax$

$$339 \quad \int \operatorname{sen} ax \, dx = -\frac{\cos ax}{a}$$

$$340 \quad \int x \operatorname{sen} ax \, dx = \frac{\operatorname{sen} ax}{a^2} - \frac{x \cos ax}{a}$$

$$341 \quad \int x^2 \operatorname{sen} ax \, dx = \frac{2x}{a^2} \operatorname{sen} ax + \left(\frac{2}{a^3} - \frac{x^2}{a} \right) \cos ax$$

$$342 \quad \int x^3 \operatorname{sen} ax \, dx = \left(\frac{3x^2}{a^2} - \frac{6}{a^4} \right) \operatorname{sen} ax + \left(\frac{6x}{a^3} - \frac{x^3}{a} \right) \cos ax$$

$$343 \quad \int \frac{\operatorname{sen} ax}{x} dx = ax - \frac{(ax)^3}{3 \cdot 3!} + \frac{(ax)^5}{5 \cdot 5!} - \dots$$

$$344 \quad \int \frac{\operatorname{sen} ax}{x^2} dx = -\frac{\operatorname{sen} ax}{x} + a \int \frac{\cos ax}{x} dx \quad \text{véase } 373$$

$$345 \quad \int \frac{dx}{\operatorname{sen} ax} = \frac{1}{a} \ln (\csc ax - \cot ax) = \frac{1}{a} \ln \tan \frac{ax}{2}$$

$$346 \quad \int \frac{x dx}{\operatorname{sen} ax} = \frac{1}{a^2} \left\{ ax + \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} + \dots + \frac{2(2^{2n-1}-1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$347 \quad \int \operatorname{sen}^2 ax \, dx = \frac{x}{2} - \frac{\operatorname{sen} 2ax}{4a}$$

- 348 $\int x \operatorname{sen}^2 ax \, dx = \frac{x^2}{4} - \frac{x \operatorname{sen} 2ax}{4a} - \frac{\cos 2ax}{8a^2}$
- 349 $\int \operatorname{sen}^3 ax \, dx = -\frac{\cos ax}{a} + \frac{\cos^3 ax}{3a}$
- 350 $\int \operatorname{sen}^4 ax \, dx = \frac{3x}{8} - \frac{\operatorname{sen} 2ax}{4a} + \frac{\operatorname{sen} 4ax}{32a}$
- 351 $\int \frac{dx}{\operatorname{sen}^2 ax} = -\frac{1}{a} \cot ax$
- 352 $\int \frac{dx}{\operatorname{sen}^3 ax} = -\frac{\cos ax}{2a \operatorname{sen}^2 ax} + \frac{1}{2a} \ln \tan \frac{ax}{2}$
- 353 $\int \operatorname{sen} px \operatorname{sen} qx \, dx = \frac{\operatorname{sen}(p-q)x}{2(p-q)} - \frac{\operatorname{sen}(p+q)x}{2(p+q)}$ Si $p = \pm q$, véase 368
- 354 $\int \frac{dx}{1 - \operatorname{sen} ax} = \frac{1}{a} \tan \left(\frac{\pi}{4} + \frac{ax}{2} \right)$
- 355 $\int \frac{x \, dx}{1 - \operatorname{sen} ax} = \frac{x}{a} \tan \left(\frac{\pi}{4} + \frac{ax}{2} \right) + \frac{2}{a^2} \ln \operatorname{sen} \left(\frac{\pi}{4} - \frac{ax}{2} \right)$
- 356 $\int \frac{dx}{1 + \operatorname{sen} ax} = -\frac{1}{a} \tan \left(\frac{\pi}{4} - \frac{ax}{2} \right)$
- 357 $\int \frac{x \, dx}{1 + \operatorname{sen} ax} = -\frac{x}{a} \tan \left(\frac{\pi}{4} - \frac{ax}{2} \right) + \frac{2}{a^2} \ln \operatorname{sen} \left(\frac{\pi}{4} + \frac{ax}{2} \right)$
- 358 $\int \frac{dx}{(1 - \operatorname{sen} ax)^2} = \frac{1}{2a} \tan \left(\frac{\pi}{4} + \frac{ax}{2} \right) + \frac{1}{6a} \tan^3 \left(\frac{\pi}{4} + \frac{ax}{2} \right)$
- 359 $\int \frac{dx}{(1 + \operatorname{sen} ax)^2} = -\frac{1}{2a} \tan \left(\frac{\pi}{4} - \frac{ax}{2} \right) - \frac{1}{6a} \tan^3 \left(\frac{\pi}{4} - \frac{ax}{2} \right)$
- 360 $\int \frac{dx}{p + q \operatorname{sen} ax} = \begin{cases} \frac{2}{a\sqrt{p^2 - q^2}} \tan^{-1} \frac{p \tan \frac{1}{2} ax + q}{\sqrt{p^2 - q^2}} \\ \frac{1}{a\sqrt{q^2 - p^2}} \ln \left(\frac{p \tan \frac{1}{2} ax + q - \sqrt{q^2 - p^2}}{p \tan \frac{1}{2} ax + q + \sqrt{q^2 - p^2}} \right) \end{cases}$
- Si $p = \pm q$ véanse 354 y 356.
- 361 $\int \frac{dx}{(p + q \operatorname{sen} ax)^2} = \frac{q \cos ax}{a(p^2 - q^2)(p + q \operatorname{sen} ax)} + \frac{p}{p^2 - q^2} \int \frac{dx}{p + q \operatorname{sen} ax}$
- Si $p = \pm q$ véanse 358 y 359.
- 362 $\int \frac{dx}{p^2 + q^2 \operatorname{sen}^2 ax} = \frac{1}{ap\sqrt{p^2 + q^2}} \tan^{-1} \frac{\sqrt{p^2 + q^2} \tan ax}{p}$
- 363 $\int \frac{dx}{p^2 - q^2 \operatorname{sen}^2 ax} = \begin{cases} \frac{1}{ap\sqrt{p^2 - q^2}} \tan^{-1} \frac{\sqrt{p^2 - q^2} \tan ax}{p} \\ \frac{1}{2ap\sqrt{q^2 - p^2}} \ln \left(\frac{\sqrt{q^2 - p^2} \tan ax + p}{\sqrt{q^2 - p^2} \tan ax - p} \right) \end{cases}$
- 364 $\int x^m \operatorname{sen} ax \, dx = -\frac{x^m \cos ax}{a} + \frac{mx^{m-1} \operatorname{sen} ax}{a^2} - \frac{m(m-1)}{a^2} \int x^{m-2} \operatorname{sen} ax \, dx$
- 365 $\int \frac{\operatorname{sen} ax}{x^n} dx = -\frac{\operatorname{sen} ax}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{\cos ax}{x^{n-1}} dx$ véase 395
- 366 $\int \operatorname{sen}^n ax \, dx = -\frac{\operatorname{sen}^{n-1} ax \cos ax}{an} + \frac{n-1}{n} \int \operatorname{sen}^{n-2} ax \, dx$
- 367 $\int \frac{dx}{\operatorname{sen}^n ax} = \frac{-\cos ax}{a(n-1) \operatorname{sen}^{n-1} ax} + \frac{n-2}{n-1} \int \frac{dx}{\operatorname{sen}^{n-2} ax}$
- 368 $\int \frac{x \, dx}{\operatorname{sen}^n ax} = \frac{-x \cos ax}{a(n-1) \operatorname{sen}^{n-1} ax} - \frac{1}{a^2(n-1)(n-2) \operatorname{sen}^{n-2} ax} + \frac{n-2}{n-1} \int \frac{x \, dx}{\operatorname{sen}^{n-2} ax}$

- 369 $\int \cos ax \, dx = \frac{\operatorname{sen} ax}{a}$
- 370 $\int x \cos ax \, dx = \frac{\cos ax}{a^2} + \frac{x \operatorname{sen} ax}{a}$
- 371 $\int x^2 \cos ax \, dx = \frac{2x}{a^2} \cos ax + \left(\frac{x^2}{a} - \frac{2}{a^3} \right) \operatorname{sen} ax$
- 372 $\int x^3 \cos ax \, dx = \left(\frac{3x^2}{a^2} - \frac{6}{a^4} \right) \cos ax + \left(\frac{x^3}{a} - \frac{6x}{a^3} \right) \operatorname{sen} ax$
- 373 $\int \frac{\cos ax}{x} \, dx = \ln x - \frac{(ax)^2}{2 \cdot 2!} + \frac{(ax)^4}{4 \cdot 4!} - \frac{(ax)^6}{6 \cdot 6!} + \dots$
- 374 $\int \frac{\cos ax}{x^2} \, dx = -\frac{\cos ax}{x} - a \int \frac{\operatorname{sen} ax}{x} \, dx$ Véase 343
- 375 $\int \frac{dx}{\cos ax} = \frac{1}{a} \ln (\sec ax + \tan ax) = \frac{1}{a} \ln \tan \left(\frac{\pi}{4} + \frac{ax}{2} \right)$
- 376 $\int \frac{x \, dx}{\cos ax} = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} + \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots + \frac{E_n(ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$
- 377 $\int \cos^2 ax \, dx = \frac{x}{2} + \frac{\operatorname{sen} 2ax}{4a}$
- 378 $\int x \cos^2 ax \, dx = \frac{x^2}{4} + \frac{x \operatorname{sen} 2ax}{4a} + \frac{\cos 2ax}{8a^2}$
- 379 $\int \cos^3 ax \, dx = \frac{\operatorname{sen} ax}{a} - \frac{\operatorname{sen}^3 ax}{3a}$
- 380 $\int \cos^4 ax \, dx = \frac{3x}{8} + \frac{\operatorname{sen} 2ax}{4a} + \frac{\operatorname{sen} 4ax}{32a}$
- 381 $\int \frac{dx}{\cos^2 ax} = \frac{\tan ax}{a}$
- 382 $\int \frac{dx}{\cos^3 ax} = \frac{\operatorname{sen} ax}{2a \cos^2 ax} + \frac{1}{2a} \ln \tan \left(\frac{\pi}{4} + \frac{ax}{2} \right)$
- 383 $\int \cos ax \cos px \, dx = \frac{\operatorname{sen}(a-p)x}{2(a-p)} + \frac{\operatorname{sen}(a+p)x}{2(a+p)}$ Si $a = \pm p$, véase 377
- 384 $\int \frac{dx}{1 - \cos ax} = -\frac{1}{a} \cot \frac{ax}{2}$
- 385 $\int \frac{x \, dx}{1 - \cos ax} = -\frac{x}{a} \cot \frac{ax}{2} + \frac{2}{a^2} \ln \operatorname{sen} \frac{ax}{2}$
- 386 $\int \frac{dx}{1 + \cos ax} = \frac{1}{a} \tan \frac{ax}{2}$
- 387 $\int \frac{x \, dx}{1 + \cos ax} = \frac{x}{a} \tan \frac{ax}{2} + \frac{2}{a^2} \ln \cos \frac{ax}{2}$
- 388 $\int \frac{dx}{(1 - \cos ax)^2} = -\frac{1}{2a} \cot \frac{ax}{2} - \frac{1}{6a} \cot^3 \frac{ax}{2}$
- 389 $\int \frac{dx}{(1 + \cos ax)^2} = \frac{1}{2a} \tan \frac{ax}{2} + \frac{1}{6a} \tan^3 \frac{ax}{2}$

$$390 \quad \int \frac{dx}{p + q \cos ax} = \begin{cases} \frac{2}{a\sqrt{p^2 - q^2}} \tan^{-1} \sqrt{(p-q)/(p+q)} \tan \frac{1}{2} ax \\ \frac{1}{a\sqrt{q^2 - p^2}} \ln \left(\frac{\tan \frac{1}{2} ax + \sqrt{(q+p)/(q-p)}}{\tan \frac{1}{2} ax - \sqrt{(q+p)/(q-p)}} \right) \end{cases} \quad \begin{array}{l} \text{Si } p = \pm q \text{ véanse} \\ 384 \text{ y } 386 \end{array}$$

$$391 \quad \int \frac{dx}{(p + q \cos ax)^2} = \frac{q \operatorname{sen} ax}{a(q^2 - p^2)(p + q \cos ax)} - \frac{p}{q^2 - p^2} \int \frac{dx}{p + q \cos ax} \quad \begin{array}{l} \text{Si } p = \pm q \text{ véanse} \\ 388 \text{ y } 389 \end{array}$$

$$392 \quad \int \frac{dx}{p^2 + q^2 \cos^2 ax} = \frac{1}{ap\sqrt{p^2 + q^2}} \tan^{-1} \frac{p \tan ax}{\sqrt{p^2 + q^2}}$$

$$393 \quad \int \frac{dx}{p^2 - q^2 \cos^2 ax} = \begin{cases} \frac{1}{ap\sqrt{p^2 - q^2}} \tan^{-1} \frac{p \tan ax}{\sqrt{p^2 - q^2}} \\ \frac{1}{2ap\sqrt{q^2 - p^2}} \ln \left(\frac{p \tan ax - \sqrt{q^2 - p^2}}{p \tan ax + \sqrt{q^2 - p^2}} \right) \end{cases}$$

$$394 \quad \int x^m \cos ax \, dx = \frac{x^m \operatorname{sen} ax}{a} + \frac{mx^{m-1}}{a^2} \cos ax - \frac{m(m-1)}{a^2} \int x^{m-2} \cos ax \, dx$$

$$395 \quad \int \frac{\cos ax}{x^n} \, dx = -\frac{\cos ax}{(n-1)x^{n-1}} - \frac{a}{n-1} \int \frac{\operatorname{sen} ax}{x^{n-1}} \, dx \quad \text{Véase } 365$$

$$396 \quad \int \cos^n ax \, dx = \frac{\operatorname{sen} ax \cos^{n-1} ax}{an} + \frac{n-1}{n} \int \cos^{n-2} ax \, dx$$

$$397 \quad \int \frac{dx}{\cos^n ax} = \frac{\operatorname{sen} ax}{a(n-1) \cos^{n-1} ax} + \frac{n-2}{n-1} \int \frac{dx}{\cos^{n-2} ax}$$

$$398 \quad \int \frac{x \, dx}{\cos^n ax} = \frac{x \operatorname{sen} ax}{a(n-1) \cos^{n-1} ax} - \frac{1}{a^2(n-1)(n-2) \cos^{n-2} ax} + \frac{n-2}{n-1} \int \frac{x \, dx}{\cos^{n-2} ax}$$

INTEGRALES DEL TIPO $\operatorname{sen} ax$ y $\cos ax$

$$399 \quad \int \operatorname{sen} ax \cos ax \, dx = \frac{\operatorname{sen}^2 ax}{2a}$$

$$400 \quad \int \operatorname{sen} px \cos qx \, dx = -\frac{\cos(p-q)x}{2(p-q)} - \frac{\cos(p+q)x}{2(p+q)}$$

$$401 \quad \int \operatorname{sen}^n ax \cos ax \, dx = \frac{\operatorname{sen}^{n+1} ax}{(n+1)a} \quad \text{Si } n = -1, \text{ véase } 440$$

$$402 \quad \int \cos^n ax \operatorname{sen} ax \, dx = -\frac{\cos^{n+1} ax}{(n+1)a} \quad \text{Si } n = -1, \text{ véase } 429$$

$$403 \quad \int \operatorname{sen}^2 ax \cos^2 ax \, dx = \frac{x}{8} - \frac{\operatorname{sen} 4ax}{32a}$$

$$404 \quad \int \frac{dx}{\operatorname{sen} ax \cos ax} = \frac{1}{a} \ln \tan ax$$

$$405 \quad \int \frac{dx}{\operatorname{sen}^2 ax \cos ax} = \frac{1}{a} \ln \tan \left(\frac{\pi}{4} + \frac{ax}{2} \right) - \frac{1}{a \operatorname{sen} ax}$$

$$406 \quad \int \frac{dx}{\operatorname{sen} ax \cos^2 ax} = \frac{1}{a} \ln \tan \frac{ax}{2} + \frac{1}{a \cos ax}$$

$$407 \quad \int \frac{dx}{\operatorname{sen}^2 ax \cos^2 ax} = -\frac{2 \cot 2ax}{a}$$

$$408 \quad \int \frac{\sin^2 ax}{\cos ax} dx = -\frac{\sin ax}{a} + \frac{1}{a} \ln \tan \left(\frac{ax}{2} + \frac{\pi}{4} \right)$$

$$409 \quad \int \frac{\cos^2 ax}{\sin ax} dx = \frac{\cos ax}{a} + \frac{1}{a} \ln \tan \frac{ax}{2}$$

$$410 \quad \int \frac{dx}{\cos ax(1 \pm \sin ax)} = \mp \frac{1}{2a(1 \pm \sin ax)} + \frac{1}{2a} \ln \tan \left(\frac{ax}{2} + \frac{\pi}{4} \right)$$

$$411 \quad \int \frac{dx}{\sin ax(1 \pm \cos ax)} = \pm \frac{1}{2a(1 \pm \cos ax)} + \frac{1}{2a} \ln \tan \frac{ax}{2}$$

$$412 \quad \int \frac{dx}{\sin ax \pm \cos ax} = \frac{1}{a\sqrt{2}} \ln \tan \left(\frac{ax}{2} \pm \frac{\pi}{8} \right)$$

$$413 \quad \int \frac{\sin ax dx}{\sin ax \pm \cos ax} = \frac{x}{2} \mp \frac{1}{2a} \ln (\sin ax \pm \cos ax)$$

$$414 \quad \int \frac{\cos ax dx}{\sin ax \pm \cos ax} = \pm \frac{x}{2} + \frac{1}{2a} \ln (\sin ax \pm \cos ax)$$

$$415 \quad \int \frac{\sin ax dx}{p + q \cos ax} = -\frac{1}{aq} \ln (p + q \cos ax)$$

$$416 \quad \int \frac{\cos ax dx}{p + q \sin ax} = \frac{1}{aq} \ln (p + q \sin ax)$$

$$417 \quad \int \frac{\sin ax dx}{(p + q \cos ax)^n} = \frac{1}{aq(n-1)(p + q \cos ax)^{n-1}}$$

$$418 \quad \int \frac{\cos ax dx}{(p + q \sin ax)^n} = \frac{-1}{aq(n-1)(p + q \sin ax)^{n-1}}$$

$$419 \quad \int \frac{dx}{p \sin ax + q \cos ax} = \frac{1}{a\sqrt{p^2 + q^2}} \ln \tan \left(\frac{ax + \tan^{-1}(q/p)}{2} \right)$$

$$420 \quad \int \frac{dx}{p \sin ax + q \cos ax + r} = \begin{cases} \frac{2}{a\sqrt{r^2 - p^2 - q^2}} \tan^{-1} \left(\frac{p + (r-q) \tan(ax/2)}{\sqrt{r^2 - p^2 - q^2}} \right) \\ \frac{1}{a\sqrt{p^2 + q^2 - r^2}} \ln \left(\frac{p - \sqrt{p^2 + q^2 - r^2} + (r-q) \tan(ax/2)}{p + \sqrt{p^2 + q^2 - r^2} + (r-q) \tan(ax/2)} \right) \end{cases}$$

Si $r = q$ véase 421. Si $r^2 = p^2 + q^2$ véase 422.

$$421 \quad \int \frac{dx}{p \sin ax + q(1 + \cos ax)} = \frac{1}{ap} \ln \left(q + p \tan \frac{ax}{2} \right)$$

$$422 \quad \int \frac{dx}{p \sin ax + q \cos ax \pm \sqrt{p^2 + q^2}} = \frac{-1}{a\sqrt{p^2 + q^2}} \tan \left(\frac{\pi}{4} \mp \frac{ax + \tan^{-1}(q/p)}{2} \right)$$

$$423 \quad \int \frac{dx}{p^2 \sin^2 ax + q^2 \cos^2 ax} = \frac{1}{apq} \tan^{-1} \left(\frac{p \tan ax}{q} \right)$$

$$424 \quad \int \frac{dx}{p^2 \sin^2 ax - q^2 \cos^2 ax} = \frac{1}{2apq} \ln \left(\frac{p \tan ax - q}{p \tan ax + q} \right)$$

$$425 \quad \int \sin^m ax \cos^n ax dx = \begin{cases} -\frac{\sin^{m-1} ax \cos^{n+1} ax}{a(m+n)} + \frac{m-1}{m+n} \int \sin^{m-2} ax \cos^n ax dx \\ \frac{\sin^{m+1} ax \cos^{n-1} ax}{a(m+n)} + \frac{n-1}{m+n} \int \sin^m ax \cos^{n-2} ax dx \end{cases}$$

$$426 \quad \int \frac{\operatorname{sen}^m ax}{\cos^n ax} dx = \begin{cases} \frac{\operatorname{sen}^{m-1} ax}{a(n-1) \cos^{n-1} ax} - \frac{m-1}{n-1} \int \frac{\operatorname{sen}^{m-2} ax}{\cos^{n-2} ax} dx \\ \frac{\operatorname{sen}^{m+1} ax}{a(n-1) \cos^{n-1} ax} - \frac{m-n+2}{n-1} \int \frac{\operatorname{sen}^m ax}{\cos^{n-2} ax} dx \\ \frac{-\operatorname{sen}^{m-1} ax}{a(m-n) \cos^{n-1} ax} + \frac{m-1}{m-n} \int \frac{\operatorname{sen}^{m-2} ax}{\cos^n ax} dx \end{cases}$$

$$427 \quad \int \frac{\cos^m ax}{\operatorname{sen}^n ax} dx = \begin{cases} \frac{-\cos^{m-1} ax}{a(n-1) \operatorname{sen}^{n-1} ax} - \frac{m-1}{n-1} \int \frac{\cos^{m-2} ax}{\operatorname{sen}^{n-2} ax} dx \\ \frac{-\cos^{m+1} ax}{a(n-1) \operatorname{sen}^{n-1} ax} - \frac{m-n+2}{n-1} \int \frac{\cos^m ax}{\operatorname{sen}^{n-2} ax} dx \\ \frac{\cos^{m-1} ax}{a(m-n) \operatorname{sen}^{n-1} ax} + \frac{m-1}{m-n} \int \frac{\cos^{m-2} ax}{\operatorname{sen}^n ax} dx \end{cases}$$

$$428 \quad \int \frac{dx}{\operatorname{sen}^m ax \cos^n ax} = \begin{cases} \frac{1}{a(n-1) \operatorname{sen}^{m-1} ax \cos^{n-1} ax} + \frac{m+n-2}{n-1} \int \frac{dx}{\operatorname{sen}^m ax \cos^{n-2} ax} \\ \frac{-1}{a(m-1) \operatorname{sen}^{m-1} ax \cos^{n-1} ax} + \frac{m+n-2}{m-1} \int \frac{dx}{\operatorname{sen}^{m-2} ax \cos^n ax} \end{cases}$$

INTEGRALES DEL TIPO $\tan ax$

$$429 \quad \int \tan ax \, dx = -\frac{1}{a} \ln \cos ax = \frac{1}{a} \ln \sec ax$$

$$430 \quad \int \tan^2 ax \, dx = \frac{\tan ax}{a} - x$$

$$431 \quad \int \tan^3 ax \, dx = \frac{\tan^2 ax}{2a} + \frac{1}{a} \ln \cos ax$$

$$432 \quad \int \tan^n ax \sec^2 ax \, dx = \frac{\tan^{n+1} ax}{(n+1)a}$$

$$433 \quad \int \frac{\sec^2 ax}{\tan ax} dx = \frac{1}{a} \ln \tan ax$$

$$434 \quad \int \frac{dx}{\tan ax} = \frac{1}{a} \ln \operatorname{sen} ax$$

$$435 \quad \int x \tan ax \, dx = \frac{1}{a^2} \left\{ \frac{(ax)^3}{3} + \frac{(ax)^5}{15} + \frac{2(ax)^7}{105} + \dots + \frac{2^{2n}(2^{2n}-1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$436 \quad \int \frac{\tan ax}{x} dx = ax + \frac{(ax)^3}{9} + \frac{2(ax)^5}{75} + \dots + \frac{2^{2n}(2^{2n}-1)B_n(ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$437 \quad \int x \tan^2 ax \, dx = \frac{x \tan ax}{a} + \frac{1}{a^2} \ln \cos ax - \frac{x^2}{2}$$

$$438 \quad \int \frac{dx}{p+q \tan ax} = \frac{px}{p^2+q^2} + \frac{q}{a(p^2+q^2)} \ln (q \operatorname{sen} ax + p \cos ax)$$

$$439 \quad \int \tan^n ax \, dx = \frac{\tan^{n-1} ax}{(n-1)a} - \int \tan^{n-2} ax \, dx$$

INTEGRALES DEL TIPO $\cot ax$

- 440 $\int \cot ax \, dx = \frac{1}{a} \ln \operatorname{sen} ax$
- 441 $\int \cot^2 ax \, dx = -\frac{\cot ax}{a} - x$
- 442 $\int \cot^3 ax \, dx = -\frac{\cot^2 ax}{2a} - \frac{1}{a} \ln \operatorname{sen} ax$
- 443 $\int \cot^n ax \csc^2 ax \, dx = -\frac{\cot^{n+1} ax}{(n+1)a}$
- 444 $\int \frac{\csc^2 ax}{\cot ax} \, dx = -\frac{1}{a} \ln \cot ax$
- 445 $\int \frac{dx}{\cot ax} = -\frac{1}{a} \ln \cos ax$
- 446 $\int x \cot ax \, dx = \frac{1}{a^2} \left\{ cx - \frac{(ax)^3}{9} - \frac{(ax)^5}{225} - \dots - \frac{2^{2n} B_n (ax)^{2n+1}}{(2n+1)!} - \dots \right\}$
- 447 $\int \frac{\cot ax}{x} \, dx = -\frac{1}{ax} - \frac{ax}{3} - \frac{(ax)^3}{135} - \dots - \frac{2^{2n} B_n (ax)^{2n-1}}{(2n-1)(2n)!} - \dots$
- 448 $\int x \cot^2 ax \, dx = -\frac{x \cot ax}{a} + \frac{1}{a^2} \ln \operatorname{sen} ax - \frac{x^2}{2}$
- 449 $\int \frac{dx}{p + q \cot ax} = \frac{px}{p^2 + q^2} - \frac{q}{a(p^2 + q^2)} \ln (p \operatorname{sen} ax + q \cos ax)$
- 450 $\int \cot^n ax \, dx = -\frac{\cot^{n-1} ax}{(n-1)a} - \int \cot^{n-2} ax \, dx$

INTEGRALES DEL TIPO $\sec ax$

- 451 $\int \sec ax \, dx = \frac{1}{a} \ln (\sec ax + \tan ax) = \frac{1}{a} \ln \tan \left(\frac{ax}{2} + \frac{\pi}{4} \right)$
- 452 $\int \sec^2 ax \, dx = \frac{\tan ax}{a}$
- 453 $\int \sec^3 ax \, dx = \frac{\sec ax \tan ax}{2a} + \frac{1}{2a} \ln (\sec ax + \tan ax)$
- 454 $\int \sec^n ax \tan ax \, dx = \frac{\sec^n ax}{na}$
- 455 $\int \frac{dx}{\sec ax} = \frac{\operatorname{sen} ax}{a}$
- 456 $\int x \sec ax \, dx = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} + \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots + \frac{E_n(ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$
- 457 $\int \frac{\sec ax}{x} \, dx = \ln x + \frac{(ax)^2}{4} + \frac{5(ax)^4}{96} + \frac{61(ax)^6}{4320} + \dots + \frac{E_n(ax)^{2n}}{2n(2n)!} + \dots$
- 458 $\int x \sec^2 ax \, dx = \frac{x}{a} \tan ax + \frac{1}{a^2} \ln \cos ax$

$$459 \quad \int \frac{dx}{q + p \sec ax} = \frac{x}{q} - \frac{p}{q} \int \frac{dx}{p + q \cos ax}$$

$$460 \quad \int \sec^n ax \, dx = \frac{\sec^{n-2} ax \tan ax}{a(n-1)} + \frac{n-2}{n-1} \int \sec^{n-2} ax \, dx$$

INTEGRALES DEL TIPO $\csc ax$

$$461 \quad \int \csc ax \, dx = \frac{1}{a} \ln (\csc ax - \cot ax) = \frac{1}{a} \ln \tan \frac{ax}{2}$$

$$462 \quad \int \csc^2 ax \, dx = -\frac{\cot ax}{a}$$

$$463 \quad \int \csc^3 ax \, dx = -\frac{\csc ax \cot ax}{2a} + \frac{1}{2a} \ln \tan \frac{ax}{2}$$

$$464 \quad \int \csc^n ax \cot ax \, dx = -\frac{\csc^n ax}{na}$$

$$465 \quad \int \frac{dx}{\csc ax} = -\frac{\cos ax}{a}$$

$$466 \quad \int x \csc ax \, dx = \frac{1}{a^2} \left\{ ax + \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} + \dots + \frac{2(2^{2n-1}-1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$467 \quad \int \frac{\csc ax}{x} \, dx = -\frac{1}{ax} + \frac{ax}{6} + \frac{7(ax)^3}{1080} + \dots + \frac{2(2^{2n-1}-1)B_n(ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$468 \quad \int x \csc^2 ax \, dx = -\frac{x \cot ax}{a} + \frac{1}{a^2} \ln \sin ax$$

$$469 \quad \int \frac{dx}{q + p \csc ax} = \frac{x}{q} - \frac{p}{q} \int \frac{dx}{p + q \sin ax} \quad \text{Véase 360}$$

$$470 \quad \int \csc^n ax \, dx = -\frac{\csc^{n-2} ax \cot ax}{a(n-1)} + \frac{n-2}{n-1} \int \csc^{n-2} ax \, dx$$

INTEGRALES DEL TIPO TRIGONOMETRICO RECIPROCO

$$471 \quad \int \sin^{-1} \frac{x}{a} \, dx = x \sin^{-1} \frac{x}{a} + \sqrt{a^2 - x^2}$$

$$472 \quad \int x \sin^{-1} \frac{x}{a} \, dx = \left(\frac{x^2}{2} - \frac{a^2}{4} \right) \sin^{-1} \frac{x}{a} + \frac{x\sqrt{a^2 - x^2}}{4}$$

$$473 \quad \int x^2 \sin^{-1} \frac{x}{a} \, dx = \frac{x^3}{3} \sin^{-1} \frac{x}{a} + \frac{(x^2 + 2a^2)\sqrt{a^2 - x^2}}{9}$$

$$474 \quad \int \frac{\sin^{-1}(x/a)}{x} \, dx = \frac{x}{a} + \frac{(x/a)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3(x/a)^5}{2 \cdot 4 \cdot 5 \cdot 5} + \frac{1 \cdot 3 \cdot 5(x/a)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots$$

$$475 \quad \int \frac{\sin^{-1}(x/a)}{x^2} \, dx = -\frac{\sin^{-1}(x/a)}{x} - \frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$476 \quad \int \left(\sin^{-1} \frac{x}{a} \right)^2 \, dx = x \left(\sin^{-1} \frac{x}{a} \right)^2 - 2x + 2\sqrt{a^2 - x^2} \sin^{-1} \frac{x}{a}$$

$$477 \quad \int \cos^{-1} \frac{x}{a} dx = x \cos^{-1} \frac{x}{a} - \sqrt{a^2 - x^2}$$

$$478 \quad \int x \cos^{-1} \frac{x}{a} dx = \left(\frac{x^2}{2} - \frac{a^2}{4} \right) \cos^{-1} \frac{x}{a} - \frac{x \sqrt{a^2 - x^2}}{4}$$

$$479 \quad \int x^2 \cos^{-1} \frac{x}{a} dx = \frac{x^3}{3} \cos^{-1} \frac{x}{a} - \frac{(x^2 + 2a^2) \sqrt{a^2 - x^2}}{9}$$

$$480 \quad \int \frac{\cos^{-1}(x/a)}{x} dx = \frac{\pi}{2} \ln x - \int \frac{\sin^{-1}(x/a)}{x} dx \quad \text{Véase } 474$$

$$481 \quad \int \frac{\cos^{-1}(x/a)}{x^2} dx = -\frac{\cos^{-1}(x/a)}{x} + \frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$482 \quad \int \left(\cos^{-1} \frac{x}{a} \right)^2 dx = x \left(\cos^{-1} \frac{x}{a} \right)^2 - 2x - 2\sqrt{a^2 - x^2} \cos^{-1} \frac{x}{a}$$

$$483 \quad \int \tan^{-1} \frac{x}{a} dx = x \tan^{-1} \frac{x}{a} - \frac{a}{2} \ln(x^2 + a^2)$$

$$484 \quad \int x \tan^{-1} \frac{x}{a} dx = \frac{1}{2}(x^2 + a^2) \tan^{-1} \frac{x}{a} - \frac{ax}{2}$$

$$485 \quad \int x^2 \tan^{-1} \frac{x}{a} dx = \frac{x^3}{3} \tan^{-1} \frac{x}{a} - \frac{ax^2}{6} + \frac{a^3}{6} \ln(x^2 + a^2)$$

$$486 \quad \int \frac{\tan^{-1}(x/a)}{x} dx = \frac{x}{a} - \frac{(x/a)^3}{3^2} + \frac{(x/a)^5}{5^2} - \frac{(x/a)^7}{7^2} + \dots$$

$$487 \quad \int \frac{\tan^{-1}(x/a)}{x^2} dx = -\frac{1}{x} \tan^{-1} \frac{x}{a} - \frac{1}{2a} \ln \left(\frac{x^2 + a^2}{x^2} \right)$$

$$488 \quad \int \cot^{-1} \frac{x}{a} dx = x \cot^{-1} \frac{x}{a} + \frac{a}{2} \ln(x^2 + a^2)$$

$$489 \quad \int x \cot^{-1} \frac{x}{a} dx = \frac{1}{2}(x^2 + a^2) \cot^{-1} \frac{x}{a} + \frac{ax}{2}$$

$$490 \quad \int x^2 \cot^{-1} \frac{x}{a} dx = \frac{x^3}{3} \cot^{-1} \frac{x}{a} + \frac{ax^2}{6} - \frac{a^3}{6} \ln(x^2 + a^2)$$

$$491 \quad \int \frac{\cot^{-1}(x/a)}{x} dx = \frac{\pi}{2} \ln x - \int \frac{\tan^{-1}(x/a)}{x} dx \quad \text{Véase } 486$$

$$492 \quad \int \frac{\cot^{-1}(x/a)}{x^2} dx = -\frac{\cot^{-1}(x/a)}{x} + \frac{1}{2a} \ln \left(\frac{x^2 + a^2}{x^2} \right)$$

$$493 \quad \int \sec^{-1} \frac{x}{a} dx = \begin{cases} x \sec^{-1} \frac{x}{a} - a \ln(x + \sqrt{x^2 - a^2}) & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ x \sec^{-1} \frac{x}{a} + a \ln(x + \sqrt{x^2 - a^2}) & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$494 \quad \int x \sec^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^2}{2} \sec^{-1} \frac{x}{a} - \frac{a\sqrt{x^2 - a^2}}{2} & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^2}{2} \sec^{-1} \frac{x}{a} + \frac{a\sqrt{x^2 - a^2}}{2} & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$495 \quad \int x^2 \sec^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^3}{3} \sec^{-1} \frac{x}{a} - \frac{ax\sqrt{x^2 - a^2}}{6} - \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^3}{3} \sec^{-1} \frac{x}{a} + \frac{ax\sqrt{x^2 - a^2}}{6} + \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$496 \quad \int \frac{\sec^{-1}(x/a)}{x} dx = \frac{\pi}{2} \ln x + \frac{a}{x} + \frac{(a/x)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3 (a/x)^5}{2 \cdot 4 \cdot 5 \cdot 5} + \frac{1 \cdot 3 \cdot 5 (a/x)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots$$

$$497 \quad \int \frac{\sec^{-1}(x/a)}{x^2} dx = \begin{cases} -\frac{\sec^{-1}(x/a)}{x} + \frac{\sqrt{x^2 - a^2}}{ax} & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ -\frac{\sec^{-1}(x/a)}{x} - \frac{\sqrt{x^2 - a^2}}{ax} & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$498 \quad \int \csc^{-1} \frac{x}{a} dx = \begin{cases} x \csc^{-1} \frac{x}{a} + a \ln(x + \sqrt{x^2 - a^2}) & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ x \csc^{-1} \frac{x}{a} - a \ln(x + \sqrt{x^2 - a^2}) & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$499 \quad \int x \csc^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^2}{2} \csc^{-1} \frac{x}{a} + \frac{a\sqrt{x^2 - a^2}}{2} & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^2}{2} \csc^{-1} \frac{x}{a} - \frac{a\sqrt{x^2 - a^2}}{2} & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$500 \quad \int x^2 \csc^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^3}{3} \csc^{-1} \frac{x}{a} + \frac{ax\sqrt{x^2 - a^2}}{6} + \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^3}{3} \csc^{-1} \frac{x}{a} - \frac{ax\sqrt{x^2 - a^2}}{6} - \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$501 \quad \int \frac{\csc^{-1}(x/a)}{x} dx = -\left(\frac{a}{x} + \frac{(a/x)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3 (a/x)^5}{2 \cdot 4 \cdot 5 \cdot 5} + \frac{1 \cdot 3 \cdot 5 (a/x)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots \right)$$

$$502 \quad \int \frac{\csc^{-1}(x/a)}{x^2} dx = \begin{cases} -\frac{\csc^{-1}(x/a)}{x} - \frac{\sqrt{x^2 - a^2}}{ax} & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ -\frac{\csc^{-1}(x/a)}{x} + \frac{\sqrt{x^2 - a^2}}{ax} & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$503 \quad \int x^m \sec^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \sec^{-1} \frac{x}{a} - \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{a^2 - x^2}} dx$$

$$504 \quad \int x^m \cos^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \cos^{-1} \frac{x}{a} + \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{a^2 - x^2}} dx$$

$$505 \quad \int x^m \tan^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \tan^{-1} \frac{x}{a} - \frac{a}{m+1} \int \frac{x^{m+1}}{x^2 + a^2} dx$$

$$506 \quad \int x^m \cot^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \cot^{-1} \frac{x}{a} + \frac{a}{m+1} \int \frac{x^{m+1}}{x^2 + a^2} dx$$

$$507 \quad \int x^m \sec^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^{m+1} \sec^{-1}(x/a)}{m+1} - \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^{m+1} \sec^{-1}(x/a)}{m+1} + \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$508 \quad \int x^m \csc^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^{m+1} \csc^{-1}(x/a)}{m+1} + \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^{m+1} \csc^{-1}(x/a)}{m+1} - \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

INTEGRALES DEL TIPO e^{ax}

$$509 \quad \int e^{ax} dx = \frac{e^{ax}}{a}$$

$$510 \quad \int x e^{ax} dx = \frac{e^{ax}}{a} \left(x - \frac{1}{a} \right)$$

$$511 \quad \int x^2 e^{ax} dx = \frac{e^{ax}}{a} \left(x^2 - \frac{2x}{a} + \frac{2}{a^2} \right)$$

$$512 \quad \int x^n e^{ax} dx = \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx$$

$$= \frac{e^{ax}}{a} \left(x^n - \frac{nx^{n-1}}{a} + \frac{n(n-1)x^{n-2}}{a^2} - \dots - \frac{(-1)^n n!}{a^n} \right) \quad \text{si } n = \text{entero positivo}$$

$$513 \quad \int \frac{e^{ax}}{x} dx = \ln x + \frac{ax}{1 \cdot 1!} + \frac{(ax)^2}{2 \cdot 2!} + \frac{(ax)^3}{3 \cdot 3!} + \dots$$

$$514 \quad \int \frac{e^{ax}}{x^n} dx = \frac{-e^{ax}}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{e^{ax}}{x^{n-1}} dx$$

$$515 \quad \int \frac{dx}{p + qe^{ax}} = \frac{x}{p} - \frac{1}{ap} \ln(p + qe^{ax})$$

$$516 \quad \int \frac{dx}{(p + qe^{ax})^2} = \frac{x}{p^2} + \frac{1}{ap(p + qe^{ax})} - \frac{1}{ap^2} \ln(p + qe^{ax})$$

$$517 \quad \int \frac{dx}{pe^{ax} + qe^{-ax}} = \begin{cases} \frac{1}{a\sqrt{pq}} \tan^{-1} \left(\sqrt{\frac{p}{q}} e^{ax} \right) \\ \frac{1}{2a\sqrt{-pq}} \ln \left(\frac{e^{ax} - \sqrt{-q/p}}{e^{ax} + \sqrt{-q/p}} \right) \end{cases}$$

$$518 \quad \int e^{ax} \sin bx dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2}$$

$$519 \quad \int e^{ax} \cos bx dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2}$$

$$520 \quad \int x e^{ax} \sin bx dx = \frac{x e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} - \frac{e^{ax}\{(a^2 - b^2) \sin bx - 2ab \cos bx\}}{(a^2 + b^2)^2}$$

$$521 \quad \int x e^{ax} \cos bx dx = \frac{x e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} - \frac{e^{ax}\{(a^2 - b^2) \cos bx + 2ab \sin bx\}}{(a^2 + b^2)^2}$$

$$522 \quad \int e^{ax} \ln x dx = \frac{e^{ax} \ln x}{a} - \frac{1}{a} \int \frac{e^{ax}}{x} dx$$

$$523 \quad \int e^{ax} \sin^n bx dx = \frac{e^{ax} \sin^{n-1} bx}{a^2 + n^2 b^2} (a \sin bx - nb \cos bx) + \frac{n(n-1)b^2}{a^2 + n^2 b^2} \int e^{ax} \sin^{n-2} bx dx$$

$$524 \quad \int e^{ax} \cos^n bx dx = \frac{e^{ax} \cos^{n-1} bx}{a^2 + n^2 b^2} (a \cos bx + nb \sin bx) + \frac{n(n-1)b^2}{a^2 + n^2 b^2} \int e^{ax} \cos^{n-2} bx dx$$

INTEGRALES DEL TIPO

 $\ln x$

$$525 \quad \int \ln x \, dx = x \ln x - x$$

$$526 \quad \int x \ln x \, dx = \frac{x^2}{2} (\ln x - \frac{1}{2})$$

$$527 \quad \int x^m \ln x \, dx = \frac{x^{m+1}}{m+1} \left(\ln x - \frac{1}{m+1} \right) \quad \text{Si } m = -1 \text{ véase } 528$$

$$528 \quad \int \frac{\ln x}{x} \, dx = \frac{1}{2} \ln^2 x$$

$$529 \quad \int \frac{\ln x}{x^2} \, dx = -\frac{\ln x}{x} - \frac{1}{x}$$

$$530 \quad \int \ln^2 x \, dx = x \ln^2 x - 2x \ln x + 2x$$

$$531 \quad \int \frac{\ln^n x}{x} \, dx = \frac{\ln^{n+1} x}{n+1} \quad \text{Si } n = -1 \text{ véase } 532.$$

$$532 \quad \int \frac{dx}{x \ln x} = \ln(\ln x)$$

$$533 \quad \int \frac{dx}{\ln x} = \ln(\ln x) + \ln x + \frac{\ln^2 x}{2 \cdot 2!} + \frac{\ln^3 x}{3 \cdot 3!} + \dots$$

$$534 \quad \int \frac{x^m dx}{\ln x} = \ln(\ln x) + (m+1) \ln x + \frac{(m+1)^2 \ln^2 x}{2 \cdot 2!} + \frac{(m+1)^3 \ln^3 x}{3 \cdot 3!} + \dots$$

$$535 \quad \int \ln^n x \, dx = x \ln^n x - n \int \ln^{n-1} x \, dx$$

$$536 \quad \int x^m \ln^n x \, dx = \frac{x^{m+1} \ln^n x}{m+1} - \frac{n}{m+1} \int x^m \ln^{n-1} x \, dx$$

Si $m = -1$ véase 531.

$$537 \quad \int \ln(x^2 + a^2) \, dx = x \ln(x^2 + a^2) - 2x + 2a \tan^{-1} \frac{x}{a}$$

$$538 \quad \int \ln(x^2 - a^2) \, dx = x \ln(x^2 - a^2) - 2x + a \ln \left(\frac{x+a}{x-a} \right)$$

$$539 \quad \int x^m \ln(x^2 \pm a^2) \, dx = \frac{x^{m+1} \ln(x^2 \pm a^2)}{m+1} - \frac{2}{m+1} \int \frac{x^{m+2}}{x^2 \pm a^2} \, dx$$

INTEGRALES DEL TIPO

 $\sinh ax$

$$540 \quad \int \sinh ax \, dx = \frac{\cosh ax}{a}$$

$$541 \quad \int x \sinh ax \, dx = \frac{x \cosh ax}{a} - \frac{\sinh ax}{a^2}$$

$$542 \quad \int x^2 \sinh ax \, dx = \left(\frac{x^2}{a} + \frac{2}{a^3} \right) \cosh ax - \frac{2x}{a^2} \sinh ax$$

$$543 \quad \int \frac{\sinh ax}{x} dx = ax + \frac{(ax)^3}{3 \cdot 3!} + \frac{(ax)^5}{5 \cdot 5!} + \dots$$

$$544 \quad \int \frac{\sinh ax}{x^2} dx = -\frac{\sinh ax}{x} + a \int \frac{\cosh ax}{x} dx \quad \text{Véase } 565$$

$$545 \quad \int \frac{dx}{\sinh ax} = \frac{1}{a} \ln \tanh \frac{ax}{2}$$

$$546 \quad \int \frac{x dx}{\sinh ax} = \frac{1}{a^2} \left\{ ax - \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} - \dots + \frac{2(-1)^n(2^{2n}-1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$547 \quad \int \sinh^2 ax dx = \frac{\sinh ax \cosh ax}{2a} - \frac{x}{2}$$

$$548 \quad \int x \sinh^2 ax dx = \frac{x \sinh 2ax}{4a} - \frac{\cosh 2ax}{8a^2} - \frac{x^2}{4}$$

$$549 \quad \int \frac{dx}{\sinh^2 ax} = -\frac{\coth ax}{a}$$

$$550 \quad \int \sinh ax \sinh px dx = \frac{\sinh(a+p)x}{2(a+p)} - \frac{\sinh(a-p)x}{2(a-p)}$$

Para $a = \pm p$ véase 547.

$$551 \quad \int \sinh ax \sin px dx = \frac{a \cosh ax \sin px - p \sinh ax \cos px}{a^2 + p^2}$$

$$552 \quad \int \sinh ax \cos px dx = \frac{a \cosh ax \cos px + p \sinh ax \sin px}{a^2 + p^2}$$

$$553 \quad \int \frac{dx}{p+q \sinh ax} = \frac{1}{a\sqrt{p^2+q^2}} \ln \left(\frac{qe^{ax} + p - \sqrt{p^2+q^2}}{qe^{ax} + p + \sqrt{p^2+q^2}} \right)$$

$$554 \quad \int \frac{dx}{(p+q \sinh ax)^2} = \frac{-q \cosh ax}{a(p^2+q^2)(p+q \sinh ax)} + \frac{p}{p^2+q^2} \int \frac{dx}{p+q \sinh ax}$$

$$555 \quad \int \frac{dx}{p^2+q^2 \sinh^2 ax} = \begin{cases} \frac{1}{ap\sqrt{q^2-p^2}} \tan^{-1} \frac{\sqrt{q^2-p^2} \tanh ax}{p} \\ \frac{1}{2ap\sqrt{p^2-q^2}} \ln \left(\frac{p+\sqrt{p^2-q^2} \tanh ax}{p-\sqrt{p^2-q^2} \tanh ax} \right) \end{cases}$$

$$556 \quad \int \frac{dx}{p^2-q^2 \sinh^2 ax} = \frac{1}{2ap\sqrt{p^2+q^2}} \ln \left(\frac{p+\sqrt{p^2+q^2} \tanh ax}{p-\sqrt{p^2+q^2} \tanh ax} \right)$$

$$557 \quad \int x^m \sinh ax dx = \frac{x^m \cosh ax}{a} - \frac{m}{a} \int x^{m-1} \cosh ax dx \quad \text{Véase } 585$$

$$558 \quad \int \sinh^n ax dx = \frac{\sinh^{n-1} ax \cosh ax}{an} - \frac{n-1}{n} \int \sinh^{n-2} ax dx$$

$$559 \quad \int \frac{\sinh ax}{x^n} dx = \frac{-\sinh ax}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{\cosh ax}{x^{n-1}} dx \quad \text{Véase } 587$$

$$560 \quad \int \frac{dx}{\sinh^n ax} = \frac{-\cosh ax}{a(n-1) \sinh^{n-1} ax} - \frac{n-2}{n-1} \int \frac{dx}{\sinh^{n-2} ax}$$

$$561 \quad \int \frac{x dx}{\sinh^n ax} = \frac{-x \cosh ax}{a(n-1) \sinh^{n-1} ax} - \frac{1}{a^2(n-1)(n-2) \sinh^{n-2} ax} - \frac{n-2}{n-1} \int \frac{x dx}{\sinh^{n-2} ax}$$

$$562 \quad \int \cosh ax \, dx = \frac{\sinh ax}{a}$$

$$563 \quad \int x \cosh ax \, dx = \frac{x \sinh ax}{a} - \frac{\cosh ax}{a^2}$$

$$564 \quad \int x^2 \cosh ax \, dx = -\frac{2x \cosh ax}{a^2} + \left(\frac{x^2}{a} + \frac{2}{a^3} \right) \sinh ax$$

$$565 \quad \int \frac{\cosh ax}{x} \, dx = \ln x + \frac{(ax)^2}{2 \cdot 2!} + \frac{(ax)^4}{4 \cdot 4!} + \frac{(ax)^6}{6 \cdot 6!} + \dots$$

$$566 \quad \int \frac{\cosh ax}{x^2} \, dx = -\frac{\cosh ax}{x} + a \int \frac{\sinh ax}{x} \, dx \quad \text{Véase 543}$$

$$567 \quad \int \frac{dx}{\cosh ax} = \frac{2}{a} \tan^{-1} e^{ax}$$

$$568 \quad \int \frac{x \, dx}{\cosh ax} = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} - \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots + \frac{(-1)^n E_n (ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$$

$$569 \quad \int \cosh^2 ax \, dx = \frac{x}{2} + \frac{\sinh ax \cosh ax}{2}$$

$$570 \quad \int x \cosh^2 ax \, dx = \frac{x^2}{4} + \frac{x \sinh 2ax}{4a} - \frac{\cosh 2ax}{8a^2}$$

$$571 \quad \int \frac{dx}{\cosh^2 ax} = \frac{\tanh ax}{a}$$

$$572 \quad \int \cosh ax \cosh px \, dx = \frac{\sinh(a-p)x}{2(a-p)} + \frac{\sinh(a+p)x}{2(a+p)}$$

$$573 \quad \int \cosh ax \sin px \, dx = \frac{a \sinh ax \sin px - p \cosh ax \cos px}{a^2 + p^2}$$

$$574 \quad \int \cosh ax \cos px \, dx = \frac{a \sinh ax \cos px + p \cosh ax \sin px}{a^2 + p^2}$$

$$575 \quad \int \frac{dx}{\cosh ax + 1} = \frac{1}{a} \tanh \frac{ax}{2}$$

$$576 \quad \int \frac{dx}{\cosh ax - 1} = -\frac{1}{a} \coth \frac{ax}{2}$$

$$577 \quad \int \frac{x \, dx}{\cosh ax + 1} = \frac{x}{a} \tanh \frac{ax}{2} - \frac{2}{a^2} \ln \cosh \frac{ax}{2}$$

$$578 \quad \int \frac{x \, dx}{\cosh ax - 1} = -\frac{x}{a} \coth \frac{ax}{2} + \frac{2}{a^2} \ln \sinh \frac{ax}{2}$$

$$579 \quad \int \frac{dx}{(\cosh ax + 1)^2} = \frac{1}{2a} \tanh \frac{ax}{2} - \frac{1}{6a} \tanh^3 \frac{ax}{2}$$

$$580 \quad \int \frac{dx}{(\cosh ax - 1)^2} = \frac{1}{2a} \coth \frac{ax}{2} - \frac{1}{6a} \coth^3 \frac{ax}{2}$$

$$581 \quad \int \frac{dx}{p + q \cosh ax} = \begin{cases} \frac{2}{a\sqrt{q^2 - p^2}} \tan^{-1} \frac{qe^{ax} + p}{\sqrt{q^2 - p^2}} \\ \frac{1}{a\sqrt{p^2 - q^2}} \ln \left(\frac{qe^{ax} + p - \sqrt{p^2 - q^2}}{qe^{ax} + p + \sqrt{p^2 - q^2}} \right) \end{cases}$$

$$582 \quad \int \frac{dx}{(p + q \cosh ax)^2} = \frac{q \sinh ax}{a(q^2 - p^2)(p + q \cosh ax)} - \frac{p}{q^2 - p^2} \int \frac{dx}{p + q \cosh ax}$$

$$583 \quad \int \frac{dx}{p^2 - q^2 \cosh^2 ax} = \begin{cases} \frac{1}{2ap\sqrt{p^2 - q^2}} \ln \left(\frac{p \tanh ax + \sqrt{p^2 - q^2}}{p \tanh ax - \sqrt{p^2 - q^2}} \right) \\ \frac{-1}{ap\sqrt{q^2 - p^2}} \tan^{-1} \frac{p \tanh ax}{\sqrt{q^2 - p^2}} \end{cases}$$

$$584 \quad \int \frac{dx}{p^2 + q^2 \cosh^2 ax} = \begin{cases} \frac{1}{2ap\sqrt{p^2 + q^2}} \ln \left(\frac{p \tanh ax + \sqrt{p^2 + q^2}}{p \tanh ax - \sqrt{p^2 + q^2}} \right) \\ \frac{1}{ap\sqrt{p^2 + q^2}} \tan^{-1} \frac{p \tanh ax}{\sqrt{p^2 + q^2}} \end{cases}$$

$$585 \quad \int x^m \cosh ax \, dx = \frac{x^m \sinh ax}{a} - \frac{m}{a} \int x^{m-1} \sinh ax \, dx \quad \text{Véase } 557$$

$$586 \quad \int \cosh^n ax \, dx = \frac{\cosh^{n-1} ax \sinh ax}{an} + \frac{n-1}{n} \int \cosh^{n-2} ax \, dx$$

$$587 \quad \int \frac{\cosh ax}{x^n} \, dx = \frac{-\cosh ax}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{\sinh ax}{x^{n-1}} \, dx \quad \text{Véase } 559$$

$$588 \quad \int \frac{dx}{\cosh^n ax} = \frac{\sinh ax}{a(n-1) \cosh^{n-1} ax} + \frac{n-2}{n-1} \int \frac{dx}{\cosh^{n-2} ax}$$

$$589 \quad \int \frac{x \, dx}{\cosh^n ax} = \frac{x \sinh ax}{a(n-1) \cosh^{n-1} ax} + \frac{1}{(n-1)(n-2)a^2 \cosh^{n-2} ax} + \frac{n-2}{n-1} \int \frac{x \, dx}{\cosh^{n-2} ax}$$

INTEGRALES DEL TIPO $\sinh ax$ y $\cosh ax$

$$590 \quad \int \sinh ax \cosh ax \, dx = \frac{\sinh^2 ax}{2a}$$

$$591 \quad \int \sinh px \cosh qx \, dx = \frac{\cosh (p+q)x}{2(p+q)} + \frac{\cosh (p-q)x}{2(p-q)}$$

$$592 \quad \int \sinh^n ax \cosh ax \, dx = \frac{\sinh^{n+1} ax}{(n+1)a} \quad \text{Si } n = -1, \text{ véase } 615$$

$$593 \quad \int \cosh^n ax \sinh ax \, dx = \frac{\cosh^{n+1} ax}{(n+1)a} \quad \text{Si } n = -1, \text{ véase } 604$$

$$594 \quad \int \sinh^2 ax \cosh^2 ax \, dx = \frac{\sinh 4ax}{32a} - \frac{x}{8}$$

$$595 \quad \int \frac{dx}{\sinh ax \cosh ax} = \frac{1}{a} \ln \tanh ax$$

$$596 \quad \int \frac{dx}{\sinh^2 ax \cosh ax} = -\frac{1}{a} \tan^{-1} \sinh ax - \frac{\operatorname{csch} ax}{a}$$

$$597 \quad \int \frac{dx}{\sinh ax \cosh^2 ax} = \frac{\operatorname{sech} ax}{a} + \frac{1}{a} \ln \tanh \frac{ax}{2}$$

$$598 \quad \int \frac{dx}{\sinh^2 ax \cosh^2 ax} = -\frac{2 \coth 2ax}{a}$$

$$599 \quad \int \frac{\sinh^2 ax}{\cosh ax} \, dx = \frac{\sinh ax}{a} - \frac{1}{a} \tan^{-1} \sinh ax$$

$$600 \quad \int \frac{\cosh^2 ax}{\sinh ax} \, dx = \frac{\cosh ax}{a} + \frac{1}{a} \ln \tanh \frac{ax}{2}$$

$$601 \quad \int \frac{dx}{\cosh ax (1 + \sinh ax)} = \frac{1}{2a} \ln \left(\frac{1 + \sinh ax}{\cosh ax} \right) + \frac{1}{a} \tan^{-1} e^{ax}$$

$$602 \quad \int \frac{dx}{\sinh ax (\cosh ax + 1)} = \frac{1}{2a} \ln \tanh \frac{ax}{2} + \frac{1}{2a(\cosh ax + 1)}$$

$$603 \quad \int \frac{dx}{\sinh ax (\cosh ax - 1)} = -\frac{1}{2a} \ln \tanh \frac{ax}{2} - \frac{1}{2a(\cosh ax - 1)}$$

INTEGRALES DEL TIPO $\tanh ax$

$$604 \quad \int \tanh ax \, dx = \frac{1}{a} \ln \cosh ax$$

$$605 \quad \int \tanh^2 ax \, dx = x - \frac{\tanh ax}{a}$$

$$606 \quad \int \tanh^3 ax \, dx = \frac{1}{a} \ln \cosh ax - \frac{\tanh^2 ax}{2a}$$

$$607 \quad \int \tanh^n ax \operatorname{sech}^2 ax \, dx = \frac{\tanh^{n+1} ax}{(n+1)a}$$

$$608 \quad \int \frac{\operatorname{sech}^2 ax}{\tanh ax} \, dx = \frac{1}{a} \ln \tanh ax$$

$$609 \quad \int \frac{dx}{\tanh ax} = \frac{1}{a} \ln \sinh ax$$

$$610 \quad \int x \tanh ax \, dx = \frac{1}{a^2} \left\{ \frac{(ax)^3}{3} - \frac{(ax)^5}{15} + \frac{2(ax)^7}{105} - \dots \frac{(-1)^{n-1} 2^{2n} (2^{2n} - 1) B_n (ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$611 \quad \int x \tanh^2 ax \, dx = \frac{x^2}{2} - \frac{x \tanh ax}{a} + \frac{1}{a^2} \ln \cosh ax$$

$$612 \quad \int \frac{\tanh ax}{x} \, dx = ax - \frac{(ax)^3}{9} + \frac{2(ax)^5}{75} - \dots \frac{(-1)^{n-1} 2^{2n} (2^{2n} - 1) B_n (ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$613 \quad \int \frac{dx}{p + q \tanh ax} = \frac{px}{p^2 - q^2} - \frac{q}{a(p^2 - q^2)} \ln (q \sinh ax + p \cosh ax)$$

$$614 \quad \int \tanh^n ax \, dx = \frac{-\tanh^{n-1} ax}{a(n-1)} + \int \tanh^{n-2} ax \, dx$$

INTEGRALES DEL TIPO $\coth ax$

$$615 \quad \int \coth ax \, dx = \frac{1}{a} \ln \sinh ax$$

$$616 \quad \int \coth^2 ax \, dx = x - \frac{\coth ax}{a}$$

$$617 \quad \int \coth^3 ax \, dx = \frac{1}{a} \ln \sinh ax - \frac{\coth^2 ax}{2a}$$

$$618 \quad \int \coth^n ax \operatorname{csch}^2 ax \, dx = -\frac{\coth^{n+1} ax}{(n+1)a}$$

$$619 \quad \int \frac{\operatorname{csch}^2 ax}{\coth ax} \, dx = -\frac{1}{a} \ln \coth ax$$

$$620 \quad \int \frac{dx}{\coth ax} = \frac{1}{a} \ln \cosh ax$$

$$621 \quad \int x \coth ax \, dx = \frac{1}{a^2} \left\{ ax + \frac{(ax)^3}{9} - \frac{(ax)^5}{225} + \dots - \frac{(-1)^{n-1} 2^{2n} B_n (ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$622 \quad \int x \coth^2 ax \, dx = \frac{x^2}{2} - \frac{x \coth ax}{a} + \frac{1}{a^2} \ln \sinh ax$$

$$623 \quad \int \frac{\coth ax}{x} \, dx = -\frac{1}{ax} + \frac{ax}{3} - \frac{(ax)^3}{135} + \dots - \frac{(-1)^n 2^{2n} B_n (ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$624 \quad \int \frac{dx}{p + q \coth ax} = \frac{px}{p^2 - q^2} - \frac{q}{a(p^2 - q^2)} \ln(p \sinh ax + q \cosh ax)$$

$$625 \quad \int \coth^n ax \, dx = -\frac{\coth^{n-1} ax}{a(n-1)} + \int \coth^{n-1} ax \, dx$$

INTEGRALES DEL TIPO $\operatorname{sech} ax$

$$626 \quad \int \operatorname{sech} ax \, dx = \frac{2}{a} \tan^{-1} e^{ax}$$

$$627 \quad \int \operatorname{sech}^2 ax \, dx = \frac{\tanh ax}{a}$$

$$628 \quad \int \operatorname{sech}^3 ax \, dx = \frac{\operatorname{sech} ax \tanh ax}{2a} + \frac{1}{2a} \tan^{-1} \sinh ax$$

$$629 \quad \int \operatorname{sech}^n ax \tanh ax \, dx = -\frac{\operatorname{sech}^n ax}{na}$$

$$630 \quad \int \frac{dx}{\operatorname{sech} ax} = \frac{\sinh ax}{a}$$

$$631 \quad \int x \operatorname{sech} ax \, dx = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} - \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots - \frac{(-1)^n E_n (ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$$

$$632 \quad \int x \operatorname{sech}^2 ax \, dx = \frac{x \tanh ax}{a} - \frac{1}{a^2} \ln \cosh ax$$

$$633 \quad \int \frac{\operatorname{sech} ax}{x} \, dx = \ln x - \frac{(ax)^2}{4} + \frac{5(ax)^4}{96} - \frac{61(ax)^6}{4320} + \dots - \frac{(-1)^n E_n (ax)^{2n}}{2n(2n)!} + \dots$$

$$634 \quad \int \frac{dx}{q + p \operatorname{sech} ax} = \frac{x}{q} - \frac{p}{q} \int \frac{dx}{p + q \cosh ax} \quad \text{Véase 581}$$

$$635 \quad \int \operatorname{sech}^n ax \, dx = \frac{\operatorname{sech}^{n-2} ax \tanh ax}{a(n-1)} + \frac{n-2}{n-1} \int \operatorname{sech}^{n-2} ax \, dx$$

INTEGRALES DEL TIPO $\operatorname{csch} ax$

$$636 \quad \int \operatorname{csch} ax \, dx = \frac{1}{a} \ln \tanh \frac{ax}{2}$$

$$637 \quad \int \operatorname{csch}^2 ax \, dx = -\frac{\coth ax}{a}$$

$$638 \quad \int \operatorname{csch}^3 ax \, dx = -\frac{\operatorname{csch} ax \coth ax}{2a} - \frac{1}{2a} \ln \tanh \frac{ax}{2}$$

$$639 \quad \int \operatorname{csch}^n ax \coth ax \, dx = -\frac{\operatorname{csch}^n ax}{na}$$

$$\begin{aligned}
640 \quad \int \frac{dx}{\operatorname{csch} ax} &= \frac{1}{a} \cosh ax \\
641 \quad \int x \operatorname{csch} ax \, dx &= \frac{1}{a^2} \left\{ ax - \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} + \dots + \frac{2(-1)^n(2^{2n-1}-1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\} \\
642 \quad \int x \operatorname{csch}^2 ax \, dx &= -\frac{x \coth ax}{a} + \frac{1}{a^2} \ln \sinh ax \\
643 \quad \int \frac{\operatorname{csch} ax}{x} \, dx &= -\frac{1}{ax} - \frac{ax}{6} + \frac{7(ax)^3}{1080} + \dots - \frac{(-1)^n 2(2^{2n-1}-1)B_n(ax)^{2n-1}}{(2n-1)(2n)!} + \dots \\
644 \quad \int \frac{dx}{q + p \operatorname{csch} ax} &= \frac{x}{q} - \frac{p}{q} \int \frac{dx}{p + q \sinh ax} \quad \text{Véase 553} \\
645 \quad \int \operatorname{csch}^n ax \, dx &= \frac{-\operatorname{csch}^{n-2} ax \coth ax}{a(n-1)} - \frac{n-2}{n-1} \int \operatorname{csch}^{n-2} ax \, dx
\end{aligned}$$

FUNCIONES HIPERBOLICAS RECIPROCAS

$$\begin{aligned}
646 \quad \int \sinh^{-1} \frac{x}{a} \, dx &= x \sinh^{-1} \frac{x}{a} - \sqrt{x^2 + a^2} \\
647 \quad \int x \sinh^{-1} \frac{x}{a} \, dx &= \left(\frac{x^2}{2} + \frac{a^2}{4} \right) \sinh^{-1} \frac{x}{a} - \frac{x \sqrt{x^2 + a^2}}{4} \\
648 \quad \int x^2 \sinh^{-1} \frac{x}{a} \, dx &= \frac{x^3}{3} \sinh^{-1} \frac{x}{a} + \frac{(2a^2 - x^2) \sqrt{x^2 + a^2}}{9} \\
649 \quad \int \frac{\sinh^{-1}(x/a)}{x} \, dx &= \begin{cases} \frac{x}{a} - \frac{(x/a)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3(x/a)^5}{2 \cdot 4 \cdot 5 \cdot 5} - \frac{1 \cdot 3 \cdot 5(x/a)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots & |x| < a \\ \frac{\ln^2(2x/a)}{2} - \frac{(a/x)^2}{2 \cdot 2 \cdot 2} + \frac{1 \cdot 3(a/x)^4}{2 \cdot 4 \cdot 4 \cdot 4} - \frac{1 \cdot 3 \cdot 5(a/x)^6}{2 \cdot 4 \cdot 6 \cdot 6 \cdot 6} + \dots & x > a \\ -\frac{\ln^2(-2x/a)}{2} + \frac{(a/x)^2}{2 \cdot 2 \cdot 2} - \frac{1 \cdot 3(a/x)^4}{2 \cdot 4 \cdot 4 \cdot 4} + \frac{1 \cdot 3 \cdot 5(a/x)^6}{2 \cdot 4 \cdot 6 \cdot 6 \cdot 6} - \dots & x < -a \end{cases} \\
650 \quad \int \frac{\sinh^{-1}(x/a)}{x^2} \, dx &= -\frac{\sinh^{-1}(x/a)}{x} - \frac{1}{a} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right) \\
651 \quad \int \cosh^{-1} \frac{x}{a} \, dx &= \begin{cases} x \cosh^{-1}(x/a) - \sqrt{x^2 - a^2}, & \cosh^{-1}(x/a) > 0 \\ x \cosh^{-1}(x/a) + \sqrt{x^2 - a^2}, & \cosh^{-1}(x/a) < 0 \end{cases} \\
652 \quad \int x \cosh^{-1} \frac{x}{a} \, dx &= \begin{cases} \frac{1}{2}(2x^2 - a^2) \cosh^{-1}(x/a) - \frac{1}{2}x \sqrt{x^2 - a^2}, & \cosh^{-1}(x/a) > 0 \\ \frac{1}{2}(2x^2 - a^2) \cosh^{-1}(x/a) + \frac{1}{2}x \sqrt{x^2 - a^2}, & \cosh^{-1}(x/a) < 0 \end{cases} \\
653 \quad \int x^2 \cosh^{-1} \frac{x}{a} \, dx &= \begin{cases} \frac{1}{8}x^3 \cosh^{-1}(x/a) - \frac{1}{9}(x^2 + 2a^2) \sqrt{x^2 - a^2}, & \cosh^{-1}(x/a) > 0 \\ \frac{1}{8}x^3 \cosh^{-1}(x/a) + \frac{1}{9}(x^2 + 2a^2) \sqrt{x^2 - a^2}, & \cosh^{-1}(x/a) < 0 \end{cases} \\
654 \quad \int \frac{\cosh^{-1}(x/a)}{x} \, dx &= \pm \left[\frac{1}{2} \ln^2(2x/a) + \frac{(a/x)^2}{2 \cdot 2 \cdot 2} + \frac{1 \cdot 3(a/x)^4}{2 \cdot 4 \cdot 4 \cdot 4} + \frac{1 \cdot 3 \cdot 5(a/x)^6}{2 \cdot 4 \cdot 6 \cdot 6 \cdot 6} + \dots \right] \\
&\quad + \text{si } \cosh^{-1}(x/a) > 0, \quad - \text{si } \cosh^{-1}(x/a) < 0 \\
655 \quad \int \frac{\cosh^{-1}(x/a)}{x^2} \, dx &= -\frac{\cosh^{-1}(x/a)}{x} \mp \frac{1}{a} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right) \quad \begin{cases} [- \text{si } \cosh^{-1}(x/a) > 0, \\ + \text{si } \cosh^{-1}(x/a) < 0] \end{cases} \\
656 \quad \int \tanh^{-1} \frac{x}{a} \, dx &= x \tanh^{-1} \frac{x}{a} + \frac{a}{2} \ln(a^2 - x^2) \\
657 \quad \int x \tanh^{-1} \frac{x}{a} \, dx &= \frac{ax^2}{2} + \frac{1}{2}(x^2 - a^2) \tanh^{-1} \frac{x}{a} \\
658 \quad \int x^2 \tanh^{-1} \frac{x}{a} \, dx &= \frac{ax^2}{6} + \frac{x^3}{3} \tanh^{-1} \frac{x}{a} + \frac{a^3}{6} \ln(a^2 - x^2)
\end{aligned}$$

$$\begin{aligned}
659 \quad \int \frac{\tanh^{-1}(x/a)}{x} dx &= \frac{x}{a} + \frac{(x/a)^3}{3^2} + \frac{(x/a)^5}{5^2} + \dots \\
660 \quad \int \frac{\tanh^{-1}(x/a)}{x^2} dx &= -\frac{\tanh^{-1}(x/a)}{x} + \frac{1}{2a} \ln \left(\frac{x^2}{a^2 - x^2} \right) \\
661 \quad \int \coth^{-1} \frac{x}{a} dx &= x \coth^{-1} x + \frac{a}{2} \ln(x^2 - a^2) \\
662 \quad \int x \coth^{-1} \frac{x}{a} dx &= \frac{ax}{2} + \frac{1}{2}(x^2 - a^2) \coth^{-1} \frac{x}{a} \\
663 \quad \int x^2 \coth^{-1} \frac{x}{a} dx &= \frac{ax^2}{6} + \frac{x^3}{3} \coth^{-1} \frac{x}{a} + \frac{a^3}{6} \ln(x^2 - a^2) \\
664 \quad \int \frac{\coth^{-1}(x/a)}{x} dx &= -\left(\frac{a}{x} + \frac{(a/x)^3}{3^2} + \frac{(a/x)^5}{5^2} + \dots \right) \\
665 \quad \int \frac{\coth^{-1}(x/a)}{x^2} dx &= -\frac{\coth^{-1}(x/a)}{x} + \frac{1}{2a} \ln \left(\frac{x^2}{x^2 - a^2} \right) \\
666 \quad \int \operatorname{sech}^{-1} \frac{x}{a} dx &= \begin{cases} x \operatorname{sech}^{-1}(x/a) + a \operatorname{sen}^{-1}(x/a), & \operatorname{sech}^{-1}(x/a) > 0 \\ x \operatorname{sech}^{-1}(x/a) - a \operatorname{sen}^{-1}(x/a), & \operatorname{sech}^{-1}(x/a) < 0 \end{cases} \\
667 \quad \int x \operatorname{sech}^{-1} \frac{x}{a} dx &= \begin{cases} \frac{1}{2} x^2 \operatorname{sech}^{-1}(x/a) - \frac{1}{2} a \sqrt{a^2 - x^2}, & \operatorname{sech}^{-1}(x/a) > 0 \\ \frac{1}{2} x^2 \operatorname{sech}^{-1}(x/a) + \frac{1}{2} a \sqrt{a^2 - x^2}, & \operatorname{sech}^{-1}(x/a) < 0 \end{cases} \\
668 \quad \int \frac{\operatorname{sech}^{-1}(x/a)}{x} dx &= \begin{cases} -\frac{1}{2} \ln(a/x) \ln(4a/x) - \frac{(x/a)^2}{2 \cdot 2 \cdot 2} - \frac{1 \cdot 3(x/a)^4}{2 \cdot 4 \cdot 4 \cdot 4} - \dots, & \operatorname{sech}^{-1}(x/a) > 0 \\ \frac{1}{2} \ln(a/x) \ln(4a/x) + \frac{(x/a)^2}{2 \cdot 2 \cdot 2} + \frac{1 \cdot 3(x/a)^4}{2 \cdot 4 \cdot 4 \cdot 4} + \dots, & \operatorname{sech}^{-1}(x/a) < 0 \end{cases} \\
669 \quad \int \operatorname{csch}^{-1} \frac{x}{a} dx &= x \operatorname{csch}^{-1} \frac{x}{a} \pm a \operatorname{senh}^{-1} \frac{x}{a} \quad [+ \text{ si } x > 0, - \text{ si } x < 0] \\
670 \quad \int x \operatorname{csch}^{-1} \frac{x}{a} dx &= \frac{x^2}{2} \operatorname{csch}^{-1} \frac{x}{a} \pm \frac{a \sqrt{x^2 + a^2}}{2} \quad [+ \text{ si } x > 0, - \text{ si } x < 0] \\
671 \quad \int \frac{\operatorname{csch}^{-1}(x/a)}{x} dx &= \begin{cases} \frac{1}{2} \ln(x/a) \ln(4a/x) + \frac{1(x/a)^2}{2 \cdot 2 \cdot 2} - \frac{1 \cdot 3(x/a)^4}{2 \cdot 4 \cdot 4 \cdot 4} + \dots & 0 < x < a \\ \frac{1}{2} \ln(-x/a) \ln(-x/4a) - \frac{(x/a)^2}{2 \cdot 2 \cdot 2} + \frac{1 \cdot 3(x/a)^4}{2 \cdot 4 \cdot 4 \cdot 4} - \dots & -a < x < 0 \\ -\frac{a}{x} + \frac{(a/x)^3}{2 \cdot 3 \cdot 3} - \frac{1 \cdot 3(a/x)^5}{2 \cdot 4 \cdot 5 \cdot 5} + \dots & |x| > a \end{cases} \\
672 \quad \int x^m \operatorname{senh}^{-1} \frac{x}{a} dx &= \frac{x^{m+1}}{m+1} \operatorname{senh}^{-1} \frac{x}{a} - \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{x^2 + a^2}} dx \\
673 \quad \int x^m \cosh^{-1} \frac{x}{a} dx &= \begin{cases} \frac{x^{m+1}}{m+1} \cosh^{-1} \frac{x}{a} - \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{x^2 - a^2}} dx & \cosh^{-1}(x/a) > 0 \\ \frac{x^{m+1}}{m+1} \cosh^{-1} \frac{x}{a} + \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{x^2 - a^2}} dx & \cosh^{-1}(x/a) < 0 \end{cases} \\
674 \quad \int x^m \tanh^{-1} \frac{x}{a} dx &= \frac{x^{m+1}}{m+1} \tanh^{-1} \frac{x}{a} - \frac{a}{m+1} \int \frac{x^{m+1}}{a^2 - x^2} dx \\
675 \quad \int x^m \coth^{-1} \frac{x}{a} dx &= \frac{x^{m+1}}{m+1} \coth^{-1} \frac{x}{a} - \frac{a}{m+1} \int \frac{x^{m+1}}{a^2 - x^2} dx \\
676 \quad \int x^m \operatorname{sech}^{-1} \frac{x}{a} dx &= \begin{cases} \frac{x^{m+1}}{m+1} \operatorname{sech}^{-1} \frac{x}{a} + \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{a^2 - x^2}} & \operatorname{sech}^{-1}(x/a) > 0 \\ \frac{x^{m+1}}{m+1} \operatorname{sech}^{-1} \frac{x}{a} - \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{a^2 - x^2}} & \operatorname{sech}^{-1}(x/a) < 0 \end{cases} \\
677 \quad \int x^m \operatorname{csch}^{-1} \frac{x}{a} dx &= \frac{x^{m+1}}{m+1} \operatorname{csch}^{-1} \frac{x}{a} \pm \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 + a^2}} \quad [+ \text{ si } x > 0, - \text{ si } x < 0]
\end{aligned}$$

